Topics in Causal Inference

S&DS 6165, Yale University, Fall 2026

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Topics in Causal Inference

S&DS 6165, Yale University, Fall 2026

Course description

This course explores modern topics in causal inference with a focus on randomized experiments and experimental design. The course develops the potential outcomes framework and builds toward advanced topics including covariate balancing, overlap conditions and positivity violations, causal inference under interference, and policy evaluation. The emphasis is on both theoretical foundations and applied practice. Causal inference from observational data is de-emphasized. Students will complete presentation and project work including both applied and theoretical replication studies.

Instructor: Prof. Johan Ugander · **Meetings:** Wednesdays, 9:25a–11:20a

Schedule

Individual lectures are under development through an instructor-directed, AI-assisted process. How these notes are made. The guest lecture and presentation weeks are shown but do not have lecture-note pages; break weeks are shown as non-meeting weeks.

Lecture	Date	Topic
L01	Sep 2	Introduction
L02	Sep 9	Basic Estimators and Experimental Pitfalls
—	Sep 16	**Guest lecture — Prof. Harsh Parikh**
L03	Sep 23	Experimental Design, Blocking, Balancing
—	Sep 30	**Presentation I**
L04	Oct 7	Positivity, Trimming, Target Populations, and Partial Identification
L05	Oct 14	Interference
—	Oct 21	**Fall break — no class**
L06	Oct 28	Estimation with Structural Models

Lecture	Date	Topic
—	Nov 4	**Presentation II**
L07	Nov 11	Policy Evaluation and Panel Data
L08	Nov 18	Advanced Estimation Topics
—	Nov 25	**Thanksgiving break — no class**
L09	Dec 2	AI for Causal Inference
—	Dec 9	**Presentation III**

How These Notes Are Made

These lecture notes for the Yale S&DS graduate course "Topics in Causal Inference" are being developed through an instructor-directed, AI-assisted workflow. This page describes the generation process.

What guides a lecture

Every lecture begins with a public `brief.md`, written by me. The brief is a concise content outline: it records the lecture's scope and emphases, a minimal literature list, and any examples, demonstrations, constraints, or open questions I choose to specify. It does not prescribe the section structure of the finished notes or pre-write assessments of the cited literature. For transparency, each lecture page links to a minimally formatted HTML rendering of the exact `brief.md` associated with that lecture.

Each lecture folder begins with only two files: the brief and a canonical `index.qmd` containing the lecture page. Assets are added only when the lecture actually needs them.

The brief is not the only instruction supplied during drafting. Four course-wide documents establish shared expectations:

- `LECTURE_SPEC.md` defines the audience, mathematical depth, and expository style.
- `WRITING_STYLE.md` distills recurring principles from my research writing and adapts them to lecture exposition, with particular emphasis on problem-driven explanation and technical intuition.
- `NOTATION.md` records notation that should remain consistent across lectures.
- `DEMO_SPEC.md` requires interactive demonstrations to have accessible, deterministic fallbacks for PDF readers.

These public files expose the durable course-specific instructions. They do not reproduce the model provider's system instructions or chain of thought during drafting. The current plan uses GPT 5.6 Sol (Extra High), with plans to experiment with the possible benefits of 5.6 Sol (Pro).

Drafting and revision process

The intended workflow is:

1. **Scope.** The instructor drafts a concise content outline and addresses lecture-specific scope, emphasis, and constraints.

2. **Literature.** The brief retains concise citations rather than extensive source commentary. Bibliographic details and technical claims are checked rather than inferred from memory. Primary papers are preferred for foundational claims and historical attributions; reviews and textbooks are used selectively when their synthesis or clarification is especially useful. Historical background and additional pointers that would interrupt the main lecture path are collected in a concise **Further reading** section.
3. **Initial generation.** OpenAI GPT-5.6 Sol, normally in either the Extra High or Pro configuration, writes an initial version directly into the lecture's `index.qmd`. Its metadata records the model and date and sets `generation-locked: true`. This flag prevents accidental wholesale regeneration; it does not make the file read-only. The model and configuration may change as the course develops, and the metadata will identify the system actually used.
4. **Brief-driven iteration.** During early drafting, the instructor may review the result, revise the brief, and explicitly request another full generation. That request temporarily clears the generation lock; the newly generated version is then locked again. A changed brief by itself never overwrites lecture text.
5. **Instructor revision.** Once the lecture's structure is satisfactory, the instructor edits the canonical lecture file directly and may substantially revise the generated material. After those edits begin, revisions are targeted so that instructor-authored text is not overwritten.
6. **Publication.** Quarto renders the same source to a web page and to PDF. Interactive web demonstrations are checked alongside their static PDF fallbacks.

Responsibility and limitations

AI assistance can accelerate outlining, exposition, examples, code, and editorial checking. It does not establish authority, verify a citation merely by naming it, or replace instructor judgment. The instructor remains responsible for what appears in the published notes.

The linked brief is best understood as the lecture's starting specification, not a complete audit log. A lecture may change after source review, classroom experience, or correction of an error. Substantive departures from the brief reflect the instructor's continuing authorship of the notes.

L01: Introduction

Potential outcomes and design-based causal inference

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Separate the causal target from choices made during design and analysis.
- Use a science table to define potential outcomes, SUTVA, and the finite-population average treatment effect.
- Define randomized designs and show how complete randomization identifies average potential outcomes and yields an unbiased estimator.

A course about choices

Causal inference begins with a comparison: what would happen under one course of action rather than another? Data do not answer that question on their own. We must decide what comparison matters, how treatment will be assigned, and how the resulting observations will be analyzed.

This course emphasizes **design-based inference**. We will treat the units and their potential outcomes as fixed and locate randomness in an assignment mechanism that we know because we chose it. That perspective keeps two kinds of choices visibly separate:

- **Design choices** determine the probability distribution used to assign treatment. Complete randomization, blocking, rerandomization, and clustered assignment are different designs.
- **Analysis choices** determine how the realized data are turned into an estimate. Difference in means, inverse-probability weighting, regression adjustment, and calibration are different analyses.

The causal target—the **estimand**—usually comes first and remains fixed while we compare designs and analyses. Sometimes the target itself changes. Trimming units with poor overlap in L04, for example, may improve the statistical problem by changing the population about which we ask the question. That is not merely a better estimator of the old target.

This distinction is a recurring discipline: **what is the question, what was randomized, and what was calculated?**

The comparison hidden in every unit

Consider six plots in an agricultural experiment. Label the plots $i = 1, \dots, 6$, and let the treatment label z equal 1 for fertilizer and 0 for control. For plot i, define $Y_i(z)$ to be the yield that would be observed if that plot were assigned treatment z. The plot therefore has two potential outcomes, $Y_i(1)$ and $Y_i(0)$, and their difference

$$\tau_i = Y_i(1) - Y_i(0)$$

is its **individual treatment effect**. Before discussing how treatment is assigned, imagine that both potential outcomes are written down for every plot.

Plot i	Yield under control $Y_i(0)$	Yield under treatment $Y_i(1)$	Individual effect $\tau_i = Y_i(1) - Y_i(0)$
1	10	13	3
2	12	12	0
3	8	12	4
4	15	14	-1
5	9	11	2
6	11	14	3

This is often called the **science table**. It is not a table we observe. Rather, it represents the fixed causal quantities that an experiment is meant to teach us about. To connect the table to observed data, we require a well-defined treatment, no interference between units, and consistency between assignment and observation; these requirements are stated more precisely below.

Let $Z_i \in \{0, 1\}$ denote plot i's actual assignment. We use an uppercase Z_i because, before randomization, the assignment is a random variable. Once the design draws its assignment, only

$$Y_i^{\text{obs}} = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$$

is observed. If $Z_i = 1$, the formula selects $Y_i(1)$; if $Z_i = 0$, it selects $Y_i(0)$. The other entry in the row remains missing. This is the fundamental obstacle: the unit-level comparison is scientifically meaningful but cannot be observed jointly for the same unit.

The table also forces a substantive question. Why summarize the six effects by their average? The average effect is often a natural policy or welfare summary, and randomization makes it especially tractable. But the table may also contain harm, unequal gains, or effects concentrated in a subgroup. An average is a target, not a law of nature.

For this course, the default target is the **finite-population average treatment effect**

$$\tau = \frac{1}{N}\sum_{i=1}^{N} \tau_i = \overline{Y}(1) - \overline{Y}(0), \tag{0.1}$$

where N is the number of experimental units and

$$\overline{Y}(z) = \frac{1}{N}\sum_{i=1}^{N} Y_i(z)$$

is the finite-population mean under treatment level z. In the table, $\overline{Y}(1) = 76/6$, $\overline{Y}(0) = 65/6$, and therefore $\tau = 11/6$. These are features of these six plots; at this point no larger population or probability model for their yields has been invoked.

What the two-column table assumes

Writing $Y_i(z)$ assumes that the single label z contains all the treatment information needed to specify unit i's potential outcome. This bundles two ideas commonly called the **stable unit treatment value assumption** (SUTVA) (Rubin 1980):

1. **No multiple versions:** the treatment labeled z has no causally distinct, unrecorded versions. If several fertilizer formulations or application protocols matter for yield, the binary label must be refined or the versions must be declared irrelevant to the question.
2. **No interference:** unit i's outcome does not depend on other units' assignments. If fertilizer runs off from a neighboring plot, $Y_i(z)$ is generally not enough; the neighbors' assignments also matter.

Consistency then links the potential outcome to observation: if the realized assignment is $Z_i = z$, then $Y_i^{\text{obs}} = Y_i(z)$. SUTVA is not a metaphysical truth. It is a simplification whose credibility depends on how units and treatments are defined. L05 will replace $Y_i(z)$ with potential outcomes indexed by richer assignment or exposure patterns. The corresponding science table becomes much larger, but the logic—define the possible interventions before asking what is identified—survives.

Neyman's move: model the assignment, not the outcomes

One response to the missing entry is to model it. We might fit a linear model, posit a distribution for outcomes, or impute each unobserved potential outcome. Such strategies can be useful, but their conclusions depend on the outcome model.

The design-based alternative starts elsewhere. Hold the entire science table fixed and introduce a known random mechanism that reveals selected entries. The probability calculation then concerns which assignment vector was drawn, not how the potential outcomes were generated.

Fisher made randomization central to experimental inference, while Neyman's agricultural experiments and survey-sampling work developed the finite-population, design-based argument (Fisher 1935; Splawa-Neyman et al. 1990; Neyman 1934). In survey sampling, known inclusion probabilities let us learn about a fixed collection of units (Horvitz and Thompson 1952). In a randomized experiment, known treatment probabilities let us learn about a fixed collection of causal contrasts. Rubin's later formulation made the potential-outcomes language central across randomized and observational studies (Rubin 1974, 2005).

The contrast with an outcome model is therefore not "statistics versus no statistics." It is a choice about what supplies the probability statement:

design-based:	condition on the potential outcomes and average over $\mathbf{Z}$,
model-based:	posit a probability law for potential outcomes or observed responses, possibly conditional on treatment and pretreatment covariates.

In the first line, repeated-assignment statements concern the same plots and the same science table under different possible treatment assignments. In the second, probability also describes how outcomes vary across units or hypothetical samples. We will often combine the two—using regression, for example, to improve precision—but the assignment mechanism will remain the foundation for causal comparisons in randomized experiments.

A randomized design is a distribution

Let

$$\mathbf{Z} = (Z_1, \dots, Z_N) \in \{0,1\}^N$$

be the assignment vector. Uppercase $\mathbf{Z}$ denotes the random vector before the design is realized; lowercase $\mathbf{z}$ denotes one particular assignment it could produce. A **randomized experimental design** is a probability distribution

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z})$$

over possible assignment vectors $\mathbf{z}$. The subscript Z emphasizes that this probability comes from the assignment mechanism, with the potential outcomes held fixed. The support

$$\mathcal{Z} = \{\mathbf{z} : \mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) > 0\}$$

records exactly which assignments the design can produce.

This definition matters. Saying "the study is randomized" is incomplete until we know both the support and the probabilities placed on it. Those choices determine which causal comparisons are possible and how estimators behave under repeated assignment.

Complete randomization

Under complete randomization, the experimenter fixes the treated count N_1 in advance and therefore also fixes the control count $N_0 = N - N_1$. Assume $0 < N_1 < N$, so both arms are nonempty. Every vector containing exactly N_1 ones is equally likely:

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) = \begin{cases} \binom{N}{N_1}^{-1}, & \sum_{i=1}^N z_i = N_1, \\ 0, & \text{otherwise.} \end{cases}$$

There are $\binom{N}{N_1}$ ways to choose the treated units, which explains the probability in the first line. Each unit has marginal treatment probability N_1/N, but the assignments are not independent. If one unit is treated, one fewer treatment slot remains. In fact, for $i \neq j$,

$$\mathbb{E}_Z(Z_i Z_j) = \mathbb{P}_Z(Z_i = 1, Z_j = 1) = \frac{N_1(N_1 - 1)}{N(N-1)},$$

and therefore

$$\mathrm{Cov}_Z(Z_i, Z_j) = -\frac{(N_1/N)(N_0/N)}{N-1}.$$

That weak negative dependence is precisely what fixes the arm sizes.

Bernoulli randomization

Under an independent Bernoulli design with common probability p,

$$Z_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p), \qquad \mathbb{P}_Z(\mathbf{Z} = \mathbf{z}) = p^{\sum_i z_i}(1-p)^{N-\sum_i z_i}.$$

Now each unit receives an independent random draw, so the treated count $N_1 = \sum_i Z_i$ is random rather than fixed by the design. Throughout the course, N_1 and N_0 denote realized arm sizes: under complete randomization their values are fixed in advance, while under Bernoulli randomization they vary across possible assignments. The Bernoulli design is operationally simple and sometimes convenient for sequential enrollment. It can also produce imbalance. The probability of an empty treatment arm is $(1-p)^N$, and the probability of an empty control arm is p^N. We return to that complication in L02.

> **i** Design comes before positivity
>
> Write $\pi_i = \mathbb{P}_Z(Z_i = 1)$ for unit i's marginal treatment probability. **Positivity** is a property of the assignment mechanism relative to the causal comparison. For both arm-specific finite-population means to be learned without extrapolation, every unit's two relevant potential outcomes must each have a chance to be observed. Thus, for the treatment-control comparison,
>
> $$0 < \pi_i < 1 \qquad \text{for every } i.$$
>
> Under nondegenerate complete randomization, $\pi_i = N_1/N$; under the common-probability Bernoulli design, $\pi_i = p$. Notice that complete randomization satisfies this unit-level requirement even though it assigns probability zero to every vector with the wrong treated count. For more complicated exposures, the relevant requirement concerns the exposure contrasts defining the estimand—not an abstract rule that every assignment vector must be possible.

Why randomization identifies an average effect

Return to complete randomization. Here N_1 and N_0 are fixed, positive constants chosen by the design. The observed treated and control means are

$$\overline{Y}_1^{\text{obs}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i^{\text{obs}}, \qquad \overline{Y}_0^{\text{obs}} = \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i^{\text{obs}}.$$

Consistency lets us rewrite these observable quantities in terms of the science table:

$$\overline{Y}_1^{\text{obs}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i(1), \qquad \overline{Y}_0^{\text{obs}} = \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i(0).$$

For example, when $Z_i = 1$, the factor Z_i selects $Y_i(1)$; when $Z_i = 0$, that unit contributes zero to the treated total. Under complete randomization, $\mathbb{E}_Z(Z_i) = N_1/N$. The notation $\mathbb{E}_Z$ means that we average only over assignments produced by the design, holding all potential outcomes fixed. Because the arm sizes are fixed, their denominators can be taken outside the expectation:

$$\mathbb{E}_Z\left(\overline{Y}_1^{\text{obs}}\right) = \frac{1}{N_1}\sum_i \mathbb{E}_Z(Z_i)Y_i(1) = \overline{Y}(1),$$
$$\mathbb{E}_Z\left(\overline{Y}_0^{\text{obs}}\right) = \frac{1}{N_0}\sum_i \mathbb{E}_Z(1-Z_i)Y_i(0) = \overline{Y}(0).$$

The cancellation is simple but important: each unit enters the treated average with probability N_1/N, and division by the fixed treated count turns those inclusion probabilities into equal weight $1/N$ on every $Y_i(1)$. The same argument applies to control.

These equalities are the design-based **identification** step: each fixed arm-specific mean is represented as the expectation of an observable random variable under the known assignment distribution. They do not identify either missing potential outcome for an individual unit. They also show why the distinction between complete and Bernoulli randomization matters. Under Bernoulli assignment, the denominators N_1 and N_0 are random and may equal zero, so one cannot repeat this calculation by simply moving them outside the expectation. L02 handles that case directly.

The corresponding **estimator** in the realized experiment is the difference in observed means,

$$\hat{\tau}_{\text{DM}} = \overline{Y}_1^{\text{obs}} - \overline{Y}_0^{\text{obs}}. \tag{0.1}$$

Linearity of design expectation then establishes its design unbiasedness:

$$\mathbb{E}_Z(\hat{\tau}_{\text{DM}}) = \tau. \tag{0.2}$$

The equality is a fact about the design. It does not require linearity, normal outcomes, constant effects, or a random sample from a larger population. It says that if we repeatedly reassigned these same N units while leaving their science table unchanged, the estimator would average to the fixed finite-population effect.

Randomization does not literally reveal both potential outcomes for any unit. It makes the observed treatment group representative of the finite population in expectation, and likewise for the control group. Identification concerns the average; unbiasedness concerns the repeated-assignment behavior of the estimator used to estimate it.

Estimand, estimator, estimate—and population

Three words prevent a great deal of confusion:

- The **estimand** is the fixed target, here τ in Equation 0.1.
- The **estimator** is the rule that maps the random assignment and resulting observed outcomes to $\hat{\tau}_{\mathrm{DM}}$. Before assignment, its value is random under the design.
- The **estimate** is the numerical value produced by the realized assignment and outcomes.

The target population must be stated just as carefully. Our τ averages over the N experimental units. It is a **finite-population** estimand. Even if those units were sampled from a larger population, treatment randomization by itself justifies the repeated-assignment statement for this realized collection of units.

We might instead posit a population distribution P for a generic unit's potential outcomes and target the **superpopulation** ATE

$$\tau_P = \mathbb{E}_P\{Y(1) - Y(0)\}.$$

Here $\mathbb{E}_P$ averages over units generated by P, rather than over alternative treatment assignments for the same units. Moving from the realized finite-population ATE to τ_P requires a sampling, transport, or superpopulation argument in addition to treatment randomization. A conditional average treatment effect,

$$\tau(x) = \mathbb{E}_P\{Y(1) - Y(0) \mid X = x\},$$

targets units with a particular value x of pretreatment covariates X. Under interference, one may instead define a global average treatment effect comparing two population-wide assignment regimes—for example, the average outcome if everyone were assigned treatment versus the average outcome if everyone were assigned control. These targets answer different questions; random assignment in one experimental sample does not automatically identify all of them.

What to carry forward

The science table states the causal comparisons. The design states which entries can be revealed and with what probabilities. The analysis states how the revealed entries are combined. Under complete randomization, the difference in observed means is design-unbiased for the finite-population average treatment effect. Every later refinement in the course should be locatable in that three-part structure.

Advanced material

Potential outcomes as missing data

Potential outcomes invite a missing-data reading: after assignment, one entry in each row of the science table is observed and the other is missing. More explicitly, the indicator that $Y_i(1)$ is observed is Z_i, while the indicator that $Y_i(0)$ is observed is $1 - Z_i$. Rubin's missing-data framework distinguishes mechanisms according to whether missingness depends on observed or unobserved values (Rubin 1976; Seaman et al. 2013). Very roughly:

- **MCAR**: missingness is independent of observed and missing values.
- **MAR**: conditional on observed information, missingness does not depend on the missing values.
- **MNAR**: dependence on missing values remains after conditioning.

Under complete randomization, assignment probabilities do not depend on the values in the fixed science table. In that limited, design-based sense, the missingness pattern for each treatment-specific outcome resembles MCAR: which entry is revealed is randomized independently of the values themselves. The assignment indicators remain dependent across units because the treated count is fixed, so the analogy should not be read as a claim that every missingness indicator is independent. More importantly, causal inference requires the interventions and their potential outcomes to be well defined; generic missing-data notation does not supply that scientific content.

Random arm sizes under Bernoulli assignment

Under i.i.d. Bernoulli assignment, $N_1 \sim \text{Binomial}(N, p)$. The usual difference in means is undefined when $N_1 = 0$ or $N_1 = N$. Conditional on any nondegenerate realized count $N_1 = n_1$, however,

$$\mathbb{P}_Z(\mathbf{Z} = \mathbf{z} \mid N_1 = n_1) = \binom{N}{n_1}^{-1} \qquad \text{when } \sum_i z_i = n_1.$$

The conditional design is therefore complete randomization with n_1 treated units, and the difference in means is conditionally design-unbiased. This does not make the ordinary difference in means unconditionally unbiased as a universally defined estimator: it still has no value on the two empty-arm events unless an extra convention is imposed. L02 introduces Horvitz–Thompson estimation, which uses the known assignment probabilities and remains defined and unbiased without conditioning on nondegeneracy.

Exact randomization variance under complete randomization

We now derive Neyman's exact variance formula rather than simply state it. The calculation has three parts: find the randomization variance of each observed arm mean, find their covariance, and then combine those pieces. The covariance matters because complete randomization selects complementary treatment and control groups rather than two independent samples.

The difference in means varies across assignments because different units enter each arm. To describe that variation, define the finite-population variances

$$S_z^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(z) - \overline{Y}(z)\}^2, \qquad S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i - \tau)^2.$$

Here S_1^2 and S_0^2 are S_z^2 evaluated at $z = 1$ and $z = 0$. All three variances are fixed features of the science table. The only randomness below comes from the assignment.

Center the science table and recall the design moments

It will help to work with outcomes centered around their finite-population means. For this derivation only, write

$$a_i = Y_i(1) - \overline{Y}(1), \qquad b_i = Y_i(0) - \overline{Y}(0),$$

so that $\sum_i a_i = \sum_i b_i = 0$. Also define the finite-population covariance between the two columns of the science table,

$$S_{10} = \frac{1}{N-1}\sum_{i=1}^{N} a_i b_i.$$

Under complete randomization, exactly N_1 of the N units are treated. A specified unit is treated with probability N_1/N, so

$$\mathrm{Var}_Z(Z_i) = \frac{N_1}{N}\left(1 - \frac{N_1}{N}\right) = \frac{N_1 N_0}{N^2}.$$

For two distinct units i and j, the probability that both are treated is

$$\mathbb{E}_Z(Z_i Z_j) = \frac{N_1}{N}\frac{N_1 - 1}{N-1} = \frac{N_1(N_1-1)}{N(N-1)}.$$

Subtracting $\mathbb{E}_Z(Z_i)\mathbb{E}_Z(Z_j)$ gives

$$\mathrm{Cov}_Z(Z_i, Z_j) = -\frac{N_1 N_0}{N^2(N-1)}, \qquad i \neq j.$$

This negative covariance is the algebraic expression of sampling without replacement: once unit i occupies one of the fixed N_1 treatment slots, there is one fewer slot available to unit j. Treating the assignment indicators as independent would discard exactly the feature of complete randomization that produces the finite-population correction.

Variance of each arm mean

Centering the treated mean gives

$$\overline{Y}_1^{\text{obs}} - \overline{Y}(1) = \frac{1}{N_1} \sum_{i=1}^{N} Z_i a_i.$$

The finite-population mean $\overline{Y}(1)$ is fixed, so it does not affect the variance. Expanding the variance of the sum into its diagonal and off-diagonal pieces yields

$$\begin{aligned}
\operatorname{Var}_Z(\overline{Y}_1^{\text{obs}}) &= \frac{1}{N_1^2} \operatorname{Var}_Z\left(\sum_i Z_i a_i \right) \\
&= \frac{1}{N_1^2} \left\{ \sum_i \operatorname{Var}_Z(Z_i) a_i^2 + \sum_{i \neq j} \operatorname{Cov}_Z(Z_i, Z_j) a_i a_j \right\} \\
&= \frac{1}{N_1^2} \left\{ \frac{N_1 N_0}{N^2} \sum_i a_i^2 - \frac{N_1 N_0}{N^2(N-1)} \sum_{i \neq j} a_i a_j \right\}.
\end{aligned}$$

Here and below, $\sum_{i \neq j}$ runs over ordered pairs of distinct units. Thus both (i, j) and (j, i) appear.

The off-diagonal sum is not a new unknown. Because the a_i are centered,

$$0 = \left(\sum_i a_i \right)^2 = \sum_i a_i^2 + \sum_{i \neq j} a_i a_j,$$

and hence $\sum_{i \neq j} a_i a_j = -\sum_i a_i^2$. Substituting this identity, and using $\sum_i a_i^2 = (N-1) S_1^2$, gives

$$\begin{aligned}
\operatorname{Var}_Z(\overline{Y}_1^{\text{obs}}) &= \frac{1}{N_1^2} \frac{N_1 N_0}{N^2} \left(1 + \frac{1}{N-1} \right) \sum_i a_i^2 \\
&= \frac{N_0}{N N_1} S_1^2 = \left(\frac{1}{N_1} - \frac{1}{N} \right) S_1^2.
\end{aligned}$$

The factor $1/N_1 - 1/N$ is the usual sampling-without-replacement correction. By the same calculation, now treating the N_0 control units as a simple random sample from the fixed $Y_i(0)$ values,

$$\operatorname{Var}_Z(\overline{Y}_0^{\text{obs}}) = \left(\frac{1}{N_0} - \frac{1}{N} \right) S_0^2.$$

Covariance between the arm means

The two arm means are built from complementary groups, so their covariance must be included. Centering the control mean makes that dependence visible:

$$\begin{aligned}\overline{Y}_0^{\text{obs}} - \overline{Y}(0) &= \frac{1}{N_0}\sum_i (1 - Z_i) b_i \\ &= -\frac{1}{N_0}\sum_i Z_i b_i.\end{aligned}$$

The second equality uses $\sum_i b_i = 0$. Therefore,

$$\text{Cov}_Z(\overline{Y}_1^{\text{obs}}, \overline{Y}_0^{\text{obs}}) = -\frac{1}{N_1 N_0}\,\text{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right).$$

Expand the covariance inside the last expression, again separating the terms with $i = j$ from those with $i \neq j$:

$$\text{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right) = \frac{N_1 N_0}{N^2}\sum_i a_i b_i - \frac{N_1 N_0}{N^2(N-1)}\sum_{i \neq j} a_i b_j.$$

Because both sets of outcomes are centered,

$$0 = \left(\sum_i a_i\right)\left(\sum_j b_j\right) = \sum_i a_i b_i + \sum_{i \neq j} a_i b_j.$$

Thus the off-diagonal sum is $-\sum_i a_i b_i$. Using the definition of S_{10} then gives

$$\text{Cov}_Z\left(\sum_i Z_i a_i, \sum_j Z_j b_j\right) = \frac{N_1 N_0}{N} S_{10},$$

and consequently

$$\text{Cov}_Z(\overline{Y}_1^{\text{obs}}, \overline{Y}_0^{\text{obs}}) = -\frac{S_{10}}{N}.$$

The minus sign comes from writing the centered control mean as a negative sum over treatment indicators. It does not imply that this covariance must be negative: its sign also depends on the finite-population covariance S_{10} between the two potential-outcome columns.

Assemble the pieces

Since $\hat{\tau}_{\mathrm{DM}} = \overline{Y}_1^{\mathrm{obs}} - \overline{Y}_0^{\mathrm{obs}}$,

$$\begin{aligned}\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) &= \mathrm{Var}_Z(\overline{Y}_1^{\mathrm{obs}}) + \mathrm{Var}_Z(\overline{Y}_0^{\mathrm{obs}}) - 2\,\mathrm{Cov}_Z(\overline{Y}_1^{\mathrm{obs}}, \overline{Y}_0^{\mathrm{obs}}) \\ &= \left(\frac{1}{N_1} - \frac{1}{N}\right) S_1^2 + \left(\frac{1}{N_0} - \frac{1}{N}\right) S_0^2 + \frac{2S_{10}}{N}.\end{aligned}$$

The remaining step rewrites the covariance between the two columns in terms of treatment-effect heterogeneity. Since

$$\tau_i - \tau = \{Y_i(1) - \overline{Y}(1)\} - \{Y_i(0) - \overline{Y}(0)\} = a_i - b_i,$$

we have

$$\begin{aligned}S_\tau^2 &= \frac{1}{N-1}\sum_i (a_i - b_i)^2 \\ &= S_1^2 + S_0^2 - 2S_{10}.\end{aligned}$$

Substituting $2S_{10} = S_1^2 + S_0^2 - S_\tau^2$ into the preceding variance expression produces Neyman's exact formula (Splawa-Neyman et al. 1990):

$$\boxed{\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}}. \tag{0.1}$$

The first two terms are the arm-specific variance contributions before accounting for how the two potential outcomes are paired within units. The subtraction by S_τ^2/N records that pairing. If every unit has the same treatment effect, then $S_\tau^2 = 0$. With heterogeneous effects, $S_\tau^2 > 0$, so the exact variance is smaller than $S_1^2/N_1 + S_0^2/N_0$. This comparison describes the terms in the displayed decomposition; it does not say that changing a science table to create more heterogeneous effects must improve precision, since S_1^2 and S_0^2 may change as well.

A check using the six-plot science table. Suppose three of the six plots are assigned to treatment. From the complete table,

$$S_1^2 = \frac{22}{15}, \qquad S_0^2 = \frac{37}{6}, \qquad S_\tau^2 = \frac{113}{30}.$$

Neyman's formula therefore gives

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{22/15}{3} + \frac{37/6}{3} - \frac{113/30}{6} = \frac{23}{12}.$$

There are $\binom{6}{3} = 20$ possible balanced assignments. Computing the difference in means under each one and averaging its squared deviation from $\tau = 11/6$ gives the same $23/12$.

The numerical agreement is a useful check on the algebra; the formula itself applies to any fixed science table under complete randomization.

We cannot generally identify S_τ^2 from the observed experiment. Its calculation requires knowing each pair $(Y_i(1), Y_i(0))$, yet randomization reveals only one member of each pair. Equivalently, the cross-column covariance S_{10} is unknown. By contrast, provided each arm contains at least two units, its observed outcomes let us estimate S_1^2 and S_0^2. Omitting the nonnegative S_τ^2/N term therefore gives an upper bound on the exact variance; replacing S_1^2 and S_0^2 with observed within-arm sample variances produces Neyman's conservative variance estimator, developed in L02.

Further reading

For the foundations of randomized experiments, read Fisher's *Design of Experiments* alongside Neyman's agricultural-experiment analysis (Fisher 1935; Splawa-Neyman et al. 1990). Neyman's representative-method essay and Horvitz–Thompson estimation make the survey-sampling lineage explicit (Neyman 1934; Horvitz and Thompson 1952). Rubin developed the modern potential-outcomes and missing-data presentation (Rubin 1974, 1976, 2005), while his comment on Fisher randomization connects potential outcomes to sharp-null randomization tests (Rubin 1980). Holland sharpened the distinction between causal effects and statistical association (Holland 1986).

Pearl's structural approach encodes causal claims through equations and graphs rather than beginning from a randomized assignment distribution (Pearl 1995, 2009). The two frameworks overlap in many applications, but this course develops the potential-outcomes, design-based route. For later examples of targets changing with the population or treatment received, see overlap-based trimming (Crump et al. 2009) and the local average treatment effect (Angrist et al. 1996).

Lecture brief: View the source `brief.md` · How these notes are made

References

L02: Basic Estimators and Experimental Pitfalls

Weighting, normalization, regression adjustment, and calibration

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a four-unit experiment to separate an estimator's randomization variance from an estimated standard error.
- Introduce inverse-probability weighting and see why random total weight can make an unbiased estimator unstable.
- Compare the HT and Hájek variances, then develop adaptive normalization as a more flexible response to random total weight.
- Reinterpret regression adjustment, CUPED, GREG, and calibration as related control-variate strategies.

From an unbiased estimator to an uncertainty statement

Under complete randomization with fixed arm sizes N_1 and N_0, L01 established that

$$\hat{\tau}_{\mathrm{DM}} = \frac{1}{N_1}\sum_{i=1}^{N} Z_i Y_i^{\mathrm{obs}} - \frac{1}{N_0}\sum_{i=1}^{N}(1-Z_i)Y_i^{\mathrm{obs}}$$

is design-unbiased for the finite-population average treatment effect τ. Unbiasedness is a statement about the center of the estimator's randomization distribution. It does not say how dispersed that distribution is.

For fixed potential outcomes, define the finite-population variances

$$S_z^2 = \frac{1}{N-1}\sum_{i=1}^{N}\{Y_i(z) - \overline{Y}(z)\}^2, \qquad S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i - \tau)^2,$$

where $\tau_i = Y_i(1) - Y_i(0)$. These are fixed features of the science table, not variances over a model for the outcomes. Neyman's exact randomization variance is (Splawa-Neyman et al. 1990)

$$V_Z := \mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}. \tag{0.1}$$

After assignment, the observed sample variance in arm z, when $N_z > 1$, is

$$s_z^2 = \frac{1}{N_z - 1}\sum_{i:Z_i=z}\{Y_i^{\mathrm{obs}} - \overline{Y}_z^{\mathrm{obs}}\}^2.$$

Keep three objects separate:

1. V_Z is a fixed but generally unknown property of the design and science table.
2. $\widehat{V}_N = s_1^2/N_1 + s_0^2/N_0$ is a random **variance estimator**, where s_z^2 is the sample variance of the observed outcomes in arm z.
3. After assignment, $\widehat{V}_N^{\mathrm{obs}}$ is the realized **variance estimate**, and $\sqrt{\widehat{V}_N^{\mathrm{obs}}}$ is the reported standard error.

Because the individual effects are not jointly observed, S_τ^2 is generally unidentified. The Neyman estimator drops its negative contribution. Under complete randomization,

$$\mathbb{E}_Z(\hat{V}_N) - V_Z = \frac{S_\tau^2}{N} \geq 0,$$

so it is conservative. When treatment effects are constant, $S_\tau^2 = 0$ and $\hat{V}_N$ is unbiased for V_Z; this does not mean that every realized variance estimate equals the true randomization variance.

A four-unit experiment by hand

Consider the fixed science table

Unit	$Y_i(0)$	$Y_i(1)$	τ_i
1	1	3	2
2	2	3	1
3	4	5	1
4	5	9	4

The finite-population ATE is $\tau = 2$. Treat exactly two units. The six equally likely assignments give

Treated units	$\hat{\tau}_{\mathrm{DM}}$
$\{1, 2\}$	-1.5
$\{1, 3\}$	0.5
$\{1, 4\}$	3.0
$\{2, 3\}$	1.0
$\{2, 4\}$	3.5
$\{3, 4\}$	5.5

For example, if units 1 and 4 are treated, their observed treated outcomes are 3 and 9, while the observed control outcomes for units 2 and 3 are 2 and 4. The treated and control means are therefore 6 and 3, giving $\hat{\tau}_{\mathrm{DM}} = 3$. The calculation for every row uses the same fixed science table but reveals a different pair of entries from each potential-outcome column.

The six estimates average to 2. Their variance, using the randomization probabilities rather than a sample-variance denominator, is

$$\frac{1}{6}\sum_{\mathbf{z}\in\mathcal{Z}}\{\hat{\tau}_{\mathrm{DM}}(\mathbf{z}) - 2\}^2 = \frac{31}{6}.$$

The same answer follows from Equation 0.1: $S_1^2 = 8$, $S_0^2 = 10/3$, and $S_\tau^2 = 2$. If units 1 and 4 are treated, the realized within-arm variances are $s_1^2 = 18$ and $s_0^2 = 2$, so $\hat{V}_N^{\mathrm{obs}} = 10$ and the reported standard error is $\sqrt{10}$. The true randomization variance remains $31/6$; one realized variance estimate need not equal it.

This tiny example is worth reconstructing at the board because it makes the source of every probability statement visible. There is one science table, six possible assignments, and a different observed estimate under each assignment.

Inverse-probability weighting

Complete randomization gave every unit the same treatment probability. More general designs need not. An experiment might deliberately assign a scarce treatment less often to expensive units, or use strata with different treatment fractions. The units whose potential outcomes are revealed under a given treatment state would then represent unequal amounts of the full experimental population. Inverse-probability weighting corrects that unequal representation.

Let

$$\pi_i = \mathbb{P}_Z(Z_i = 1), \qquad 0 < \pi_i < 1.$$

The **Horvitz–Thompson** (HT) estimators of the two finite-population means are

$$\hat{\mu}_{1,\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N} \frac{Z_i Y_i^{\mathrm{obs}}}{\pi_i}, \qquad \hat{\mu}_{0,\mathrm{HT}} = \frac{1}{N}\sum_{i=1}^{N} \frac{(1-Z_i) Y_i^{\mathrm{obs}}}{1-\pi_i},$$

and $\hat{\tau}_{\mathrm{HT}} = \hat{\mu}_{1,\mathrm{HT}} - \hat{\mu}_{0,\mathrm{HT}}$ (Horvitz and Thompson 1952). A treated unit with $\pi_i = 0.2$, for example, receives weight $1/0.2 = 5$: it is observed under treatment only one fifth of the time across repeated assignments, so each such observation represents five assignment opportunities. Formally, each unit's weighted treatment indicator has expectation one,

$$\mathbb{E}_Z\left(\frac{Z_i}{\pi_i}\right) = 1.$$

Linearity of expectation then gives

$$\mathbb{E}_Z(\hat{\mu}_{1,\mathrm{HT}}) = \frac{1}{N}\sum_i Y_i(1) = \overline{Y}(1),$$

and similarly for control. This argument uses only the marginal inclusion probabilities; the Z_i need not be independent. Positivity is essential, because a potential outcome assigned probability zero can never receive an inverse-probability correction. The conclusion is **unconditional** design unbiasedness: it averages over every assignment in the design, including the all-treated and all-control assignments that a Bernoulli design permits.

Under complete randomization, $\pi_i = N_1/N$, so

$$\hat{\mu}_{1,\mathrm{HT}} = \frac{1}{N_1}\sum_i Z_i Y_i^{\mathrm{obs}}.$$

Thus HT and difference in means coincide. Under i.i.d. Bernoulli assignment, they do not.

The Bernoulli subtlety

With common Bernoulli probability p, the usual difference in means divides by the random counts N_1 and N_0. It is undefined when either count is zero. Conditional on any $N_1 = n_1 \in \{1, \dots, N-1\}$, however, the assignment is a complete randomization with n_1 treated units. Hence

$$\mathbb{E}_Z(\hat{\tau}_{\mathrm{DM}} \mid N_1 = n_1) = \tau.$$

It is therefore conditionally unbiased on every nondegenerate arm size, but it is not an everywhere-defined, unconditionally unbiased estimator over the original Bernoulli design. By contrast,

$$\hat{\tau}_{\mathrm{HT}} = \frac{1}{N} \sum_i \left\{ \frac{Z_i Y_i^{\mathrm{obs}}}{p} - \frac{(1 - Z_i) Y_i^{\mathrm{obs}}}{1 - p} \right\}$$

is defined and design-unbiased on the full assignment space.

> **What HT is correcting**
>
> Inverse-probability weighting gives a unit that is rarely assigned to a particular treatment state more representation when that state occurs. The identity $\mathbb{E}_Z(Z_i/\pi_i) = 1$ is the entire unbiasedness argument for the treated potential-outcome mean. The cost is that small probabilities create large and noisy weights.

Hájek normalization

HT weights have the right total **in expectation**, not in every assignment. For the treated mean, their realized sum is

$$\widehat{N}_1^w = \sum_i \frac{Z_i}{\pi_i}, \qquad \mathbb{E}_Z(\widehat{N}_1^w) = N.$$

The notation $\widehat{N}_1^w$ does not denote the number of treated units. It is the population size represented by the treated units after weighting. HT divides its weighted outcome total by the known N even when $\widehat{N}_1^w \neq N$. The Hájek estimator instead divides by the realized total weight (Hájek 1971):

$$\hat{\mu}_{1,\mathrm{H}} = \frac{\sum_i Z_i Y_i^{\mathrm{obs}}/\pi_i}{\sum_i Z_i/\pi_i}, \tag{0.1}$$

with an analogous control estimator. On an empty arm its denominator is zero. Under common-probability Bernoulli assignment, the Hájek arm means are the ordinary arm means. Under complete randomization, HT, Hájek, and difference in means all coincide.

With unequal probabilities, normalization can alter both finite-sample bias and variance; neither change has a universal sign. We compare assignment variances first, then turn to ratio bias from the random denominator.

A first variance comparison: HT versus Hájek

To see what normalization changes without carrying treatment and control notation through every line, isolate one generic finite-population mean under independent Bernoulli sampling. Let $I_i \sim \mathrm{Bernoulli}(\pi_i)$ independently reveal fixed values Y_i. The target total is $T_Y = \sum_i Y_i = N\mu$. Its inverse-probability estimate and the corresponding estimated population size are

$$\hat{T}_Y = \sum_i \frac{I_i Y_i}{\pi_i}, \qquad \widehat{N} = \sum_i \frac{I_i}{\pi_i}, \qquad \mu = \frac{1}{N}\sum_i Y_i.$$

This is the treated-arm problem with $I_i = Z_i$ and $Y_i = Y_i(1)$. For the control arm, take $I_i = 1 - Z_i$, $Y_i = Y_i(0)$, and replace π_i by $1 - \pi_i$. HT is $\hat{T}_Y/N$, whereas Hájek is $\hat{T}_Y/\widehat{N}$. Both numerators estimate the same total; only the denominator differs.

For HT, the estimation error has the exact representation

$$\hat{\mu}_{\mathrm{HT}} - \mu = \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right) Y_i.$$

Independence makes the cross-unit covariance terms vanish, giving the exact variance

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{HT}}) = \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i} Y_i^2. \tag{0.2}$$

Writing $Y_i = (Y_i - \mu) + \mu$ separates two sources of error:

$$\hat{\mu}_{\mathrm{HT}} - \mu = \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \mu) + \mu\left(\frac{\widehat{N}}{N} - 1\right). \tag{0.3}$$

The second term is produced by the random total weight. If the same constant a were added to every Y_i, the target mean would increase by a, but the HT estimation error would also acquire $a(\widehat{N}/N - 1)$. Thus HT variance can depend on the common additive baseline of the outcomes, not only on how dispersed they are around their mean.

When $\widehat{N} > 0$, Hájek divides by the realized total weight, so the same additive shift simply adds a to both the estimate and its target. Its estimation error is unchanged. The price is a random denominator, which makes an exact finite-sample variance less transparent.

The Taylor calculation shows how normalization removes the random-baseline term. Apply $f(t, n) = t/n$ to $(\hat{T}_Y, \widehat{N})$ and expand around the expectations $(T_Y, N) = (N\mu, N)$. Because the two derivatives at that point are $1/N$ and $-\mu/N$,

$$\frac{\hat{T}_Y}{\widehat{N}} \approx \mu + \frac{\hat{T}_Y - T_Y}{N} - \frac{\mu}{N}(\widehat{N} - N).$$

Substituting the two weighted totals and collecting each unit's contribution gives

$$\hat{\mu}_{\mathrm{H}} - \mu \approx \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \mu). \tag{0.4}$$

Consequently, its first-order variance is

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i}(Y_i - \mu)^2. \tag{0.5}$$

The contrast between Equation 0.2 and Equation 0.5 is now explicit: HT's exact variance involves the squared outcomes Y_i^2, whereas the first-order Hájek variance involves squared deviations $(Y_i - \mu)^2$. With common $\pi_i = p$,

$$\mathrm{Var}_Z(\hat{\mu}_{\mathrm{HT}}) - \mathrm{Var}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{1-p}{pN}\mu^2.$$

The difference is large when the outcome mean is far from zero relative to the outcomes' dispersion. Equivalently, a common additive shift changes HT's variance because random total weight multiplies that shift, while it leaves the first-order Hájek variance unchanged. This is the classic survey-sampling linearization logic (Särndal et al. 1989, 1992). The calculation isolates one arm; for the treatment-effect estimator, the treated- and control-arm variances must also be combined with their design covariance.

Ratio bias from normalization

Variance is not the entire comparison. A second-order expansion gives the approximate ratio bias

$$\mathrm{Bias}_Z(\hat{\mu}_{\mathrm{H}}) \approx \frac{\mu \,\mathrm{Var}_Z(\widehat{N}) - \mathrm{Cov}_Z(\hat{T}_Y, \widehat{N})}{N^2}.$$

Here

$$\mathrm{Var}_Z(\widehat{N}) = \sum_i \frac{1-\pi_i}{\pi_i}, \qquad \mathrm{Cov}_Z(\hat{T}_Y, \widehat{N}) = \sum_i \frac{1-\pi_i}{\pi_i} Y_i.$$

The expression is informative even before it is evaluated: ratio bias appears when fluctuations in the estimated outcome total and the estimated population size do not line up as μ times one another. For common $\pi_i = p$, the two terms cancel. This agrees with the exact fact that, conditional on any positive realized sample size, the Hájek mean is the ordinary sample mean under simple random sampling. The Taylor calculations nevertheless remain approximations to an unconditional ratio problem whose denominator can be zero. They are most informative when $\widehat{N}$ is concentrated near N. With small effective samples, extreme probabilities, or influential units, higher-order terms can matter and neither estimator has a universal MSE advantage.

HT versus Hájek in a finite population

The demonstration uses

$$Y_i(0) = b + \sigma x_i, \qquad Y_i(1) = b + \sigma x_i + \tau,$$

with finite-population x_i centered at zero. The common baseline b shifts every potential outcome by the same amount, so it changes neither the individual treatment effects nor the ATE. For an HT arm mean, however, that shift adds the random error term $b(\widehat{N}/N - 1)$. Because Hájek divides by $\widehat{N}$, the same shift simply adds b to its arm-mean estimate and does not change that estimate's variance. When $b = 0$, this first-order shift-cancellation advantage disappears; the random Hájek denominator can then produce more conditional MSE than HT's fixed denominator.

The left panel uses centered outcomes with no effect; the right introduces a large common baseline. The reversal isolates the random-total-weight mechanism in Equation 0.3.

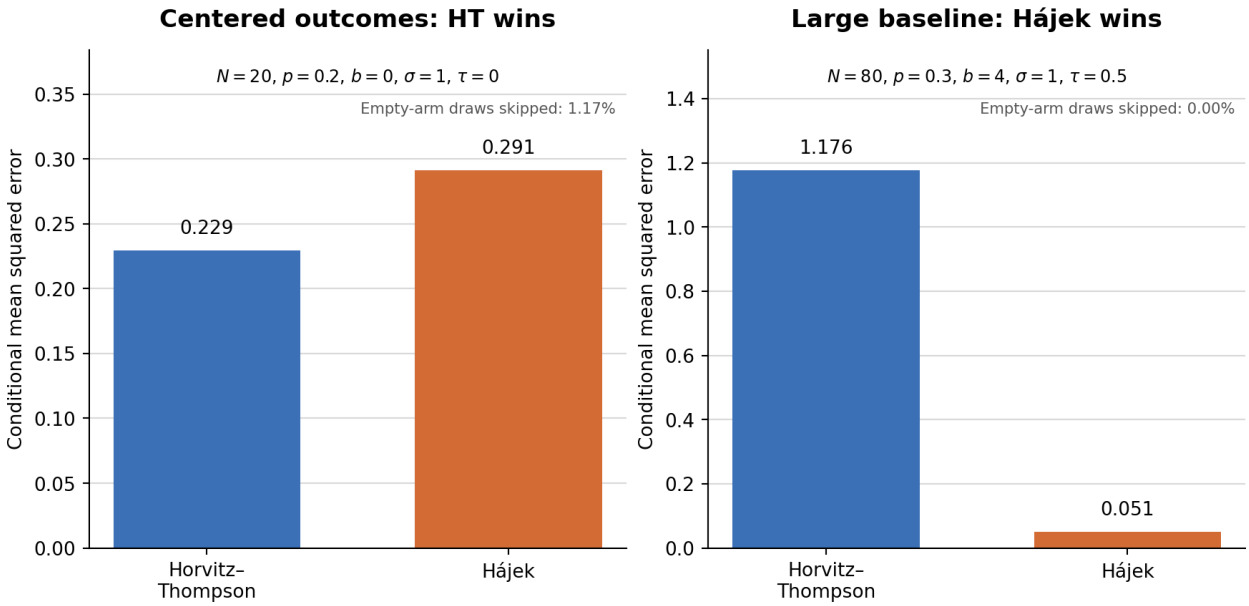


Figure 1: Two fixed simulation settings show that neither estimator uniformly minimizes finite-sample MSE. HT wins in the centered setting; Hájek wins when outcomes have a large common baseline.

Normalization is a continuum

HT leaves the realized total-weight error uncorrected; Hájek removes it completely. That contrast suggests a less binary question: how much of the denominator fluctuation should an estimator use? Trotter and Tukey introduced an affine family that makes this amount explicit; Khan and Ugander (2023) develop the family for inverse-probability weighting. For the same generic mean problem, define

$$\hat{\mu}_\lambda = \frac{\hat{T}_Y}{(1-\lambda)N + \lambda\widehat{N}} = \frac{\hat{T}_Y}{N + \lambda(\widehat{N} - N)}. \tag{0.1}$$

$\lambda = 0$ gives HT and $\lambda = 1$ gives Hájek. The second expression makes the role of λ visible: it controls how strongly the random error $\widehat{N} - N$ enters the denominator. Values outside $[0, 1]$ are allowed; the family is an affine normalization family, not a convex interpolation. In a realized finite sample, however, a chosen value is usable only when $N + \lambda(\widehat{N} - N) \neq 0$. Extrapolated values can drive this denominator near zero or make it negative, so a first-order variance optimum is not by itself a finite-sample stability guarantee.

The same first-order expansion used for Hájek yields

$$\hat{\mu}_\lambda - \mu \approx \frac{1}{N}\sum_i \left(\frac{I_i}{\pi_i} - 1\right)(Y_i - \lambda\mu). \tag{0.2}$$

This equation turns the choice of denominator into the choice of a centering constant. HT uses no centering, because $\lambda\mu = 0$. Hájek centers at the target mean, because $\lambda\mu = \mu$. An intermediate or extrapolated value uses a different constant.

Before changing probability frameworks, the optimal first-order choice can be written entirely in finite-population notation. Let $q_i = (1 - \pi_i)/\pi_i = \mathrm{Var}_Z(I_i/\pi_i)$. Under independent Bernoulli inclusion, Equation 0.2 gives

$$\mathrm{Var}_Z(\hat{\mu}_\lambda) \approx \frac{1}{N^2}\sum_i q_i(Y_i - \lambda\mu)^2.$$

When $\mu \neq 0$ and at least one $q_i > 0$, minimizing this quadratic yields

$$\lambda_N^* = \frac{\sum_i q_i Y_i}{\mu \sum_i q_i}, \qquad c_N^* := \lambda_N^*\mu = \frac{\sum_i q_i Y_i}{\sum_i q_i}. \tag{0.3}$$

Thus the useful object is not fundamentally a fraction λ. It is the centering constant c_N^*, a weighted mean of the fixed Y_i that emphasizes units with noisier inverse-probability weights.

To state the adaptive choice used by Khan and Ugander (2023), we now make an explicit change of framework. The preceding variance calculations held a finite population fixed. Their asymptotic analysis instead treats the pairs (Y_i, π_i) as i.i.d. draws of a generic pair (Y, P), where P is a random inclusion probability. In this paragraph, $\mu = \mathbb{E}(Y)$ is the superpopulation mean rather than the mean of one fixed finite population. Define the weight-noise factor $Q = (1 - P)/P$ and the two population moments

$$\Pi = \mathbb{E}(Q), \qquad A = \mathbb{E}(YQ).$$

These expectations are the superpopulation analogues of $N^{-1} \sum_i q_i$ and $N^{-1} \sum_i q_i Y_i$ in Equation 0.3.

For the usual $\sqrt{N}$ scaling, the part of the first-order asymptotic variance that depends on λ is

$$\sigma_\lambda^2 = \mathbb{E}\{Q(Y - \lambda\mu)^2\}. \tag{0.4}$$

Superpopulation sampling also contributes variation in Y itself, but that term does not depend on λ and therefore does not affect which member of the normalization family minimizes variance.

This is a quadratic in λ. When $\mu \neq 0$ and $\Pi > 0$, differentiating it gives

$$\frac{d\sigma_\lambda^2}{d\lambda} = -2\mu(A - \lambda\mu\Pi),$$

and hence the minimizer

$$\lambda^* = \frac{A}{\Pi\mu}. \tag{0.5}$$

The numerator and denominator become easier to interpret after absorbing the factor μ. Define the optimal centering constant

$$c^* = \frac{A}{\Pi}$$

so that $\lambda^* = c^*/\mu$. The constant c^* is an average of Y under weights proportional to Q: it emphasizes outcomes attached to the inclusion probabilities that create the most inverse-weight noise. If the inclusion probability is common, Q is constant, $c^* = \mu$, and Hájek is optimal within this first-order family. With heterogeneous inclusion probabilities, the high-weight portion of the population can have a mean different from μ; then neither HT nor Hájek need use the best centering constant, and λ^* need not lie between 0 and 1.

When $\mu = 0$, the product $\lambda\mu$ vanishes in Equation 0.2, so the first-order variance does not depend on λ. This does not mean that normalization is irrelevant in every finite sample. It means that λ, which divides a useful centering constant by μ, is a poor coordinate near zero. The constant c^* remains well defined.

The population moments A and Π are unknown in the generic inclusion problem. The identity $\mathbb{E}(I_i Y_i/\pi_i \mid Y_i, \pi_i) = Y_i$ motivates the IPW sample analogues

$$\widehat{A} = \frac{1}{N}\sum_i \frac{1-\pi_i}{\pi_i}\frac{I_i Y_i}{\pi_i}, \qquad \widehat{\Pi} = \frac{1}{N}\sum_i \frac{1-\pi_i}{\pi_i}\frac{I_i}{\pi_i}, \qquad \hat{c} = \frac{\widehat{A}}{\widehat{\Pi}}.$$

The remaining obstacle is μ in $\lambda^* = c^*/\mu$. Adaptive normalization uses the current mean estimate in that denominator, updates the normalization, and repeats. Starting, for example, at $\hat{\mu}^{(0)} = \hat{\mu}_{\mathrm{HT}}$, the iteration sets

$$\hat{\lambda}^{(t)} = \frac{\hat{c}}{\hat{\mu}^{(t-1)}}, \qquad \hat{\mu}^{(t)} = \frac{\hat{T}_Y}{(1-\hat{\lambda}^{(t)})N + \hat{\lambda}^{(t)}\widehat{N}}.$$

The two equations form a feedback loop: the current mean estimate supplies the denominator of $\hat{\lambda}^{(t)}$, and the resulting normalization supplies the next mean estimate. At a nonzero fixed point, substitute $\hat{\lambda} = \hat{c}/\hat{\mu}$ into the denominator and multiply through:

$$N\hat{\mu} + \hat{c}(\widehat{N} - N) = \hat{T}_Y.$$

Solving for $\hat{\mu}$ gives the closed form

$$\hat{\mu}_{\mathrm{AN}} = \frac{\hat{T}_Y}{N} + \hat{c}\left(1 - \frac{\widehat{N}}{N}\right) = \hat{\mu}_{\mathrm{HT}} - \hat{c}\left(\frac{\widehat{N}}{N} - 1\right). \tag{0.6}$$

The fixed point is therefore a **control-variate**, or regression-control, estimator. The quantity $D = \widehat{N}/N - 1$ has design expectation zero and is correlated with the part of HT error generated by total-weight fluctuation. For any *fixed* coefficient c, subtracting cD preserves exact design unbiasedness; choosing the population coefficient c^* minimizes the first-order variance in this one-dimensional control family. HT uses $c = 0$, Hájek corresponds at first order to $c = \mu$, and adaptive normalization plugs in $\hat{c}$.

Because $\hat{c}$ is estimated from the same realization as D, the mean-zero property of D alone does not prove exact unbiasedness of the adaptive estimator. The guarantee in Khan and Ugander (2023) is asymptotic: under their moment and consistency conditions, the adaptive variance is no larger than the corresponding HT or Hájek variance. Unstable estimated moments or a nearly empty effective sample can still matter in finite samples. The closed form in Equation 0.6 is also preferable if an iterate passes near zero, where the λ parameterization becomes unstable.

Regression adjustment

The weighted-population-size error is one random quantity whose expectation is known from the design. Pretreatment covariates provide many more. Suppose, for example, that the treated arm happens to contain users who were more active before an experiment. That difference cannot have been caused by treatment, but if prior activity predicts the outcome, it can make the unadjusted outcome difference unusually large or small.

Under complete randomization with fixed positive arm sizes, define the realized covariate means

$$\overline{X}_1^{\text{obs}} = \frac{1}{N_1}\sum_i Z_i X_i, \qquad \overline{X}_0^{\text{obs}} = \frac{1}{N_0}\sum_i (1 - Z_i) X_i.$$

The covariates are vectors, so these are componentwise arm averages. Their difference is observable after assignment. The superscript obs refers to the realized treatment and control groups; it does not suggest that X_i was measured after treatment. We write the difference as

$$\hat{\tau}_X = \overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}$$

This difference has design expectation zero: randomization does not guarantee balance in the realized experiment, but it makes the imbalance average to zero across assignments. If β maps a one-unit covariate imbalance into predicted outcome imbalance, the natural correction is

$$\hat{\tau}_{\text{adj}}(\beta) = \hat{\tau}_{\text{DM}} - \beta^\top \hat{\tau}_X. \tag{0.1}$$

For any coefficient β fixed before assignment, $\mathbb{E}_Z\{\hat{\tau}_{\text{adj}}(\beta)\} = \tau$ follows immediately: the adjustment subtracts a fixed multiple of a mean-zero random quantity. When X predicts outcomes, this can remove chance imbalance that would otherwise masquerade as treatment effect. An arbitrary coefficient can instead add noise; design unbiasedness and variance reduction are separate claims.

A common slope is not required. If $\overline{X} = N^{-1}\sum_i X_i$ is known, fixed treatment- and control-arm coefficients give the following estimator, where $\overline{Y}_z^{\text{obs}}$ denotes the observed outcome mean in arm z:

$$\begin{aligned}\hat{\tau}_{\text{adj}}(\beta_1, \beta_0) = &\left\{\overline{Y}_1^{\text{obs}} - (\overline{X}_1^{\text{obs}} - \overline{X})^\top \beta_1\right\} \\ &- \left\{\overline{Y}_0^{\text{obs}} - (\overline{X}_0^{\text{obs}} - \overline{X})^\top \beta_0\right\}.\end{aligned} \tag{0.2}$$

For fixed coefficients, each bracket is design-unbiased for its corresponding finite-population potential-outcome mean. Taking $\beta_1 = \beta_0 = \beta$ recovers Equation 0.1. In practice, the coefficients are usually fitted from the experimental outcomes. They are then assignment-dependent, so the exact finite-sample mean-zero argument no longer follows merely from $\mathbb{E}_Z(\hat{\tau}_X) = 0$. The regression should

allow treatment-specific slopes—equivalently, include treatment-by-centered-covariate interactions—because the two potential-outcome surfaces need not be parallel (Freedman 2008; Lin 2013). Randomization protects the large-sample causal interpretation under suitable regularity conditions, but careless specifications or standard errors can lose efficiency or misstate uncertainty.

This returns us to normalization. In Equation 0.6, $\widehat{N}/N - 1$ is the inverse-weighted imbalance in the constant feature 1: its target mean is exactly one, but its realized weighted mean need not be. Adaptive normalization uses that scalar imbalance as a control variable. Conventional regression adjustment applies the same logic to informative pretreatment features. GREG makes the connection especially explicit.

GREG from a known baseline total

Suppose an online platform fixes a population of N accounts before an experiment. For every account it already knows X_i, the number of active days during the preceding four weeks. The finite-population mean

$$\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$$

is therefore known exactly before assignment. By contrast, $Y_i(1)$, the experiment-period outcome under treatment, is observed only for treated accounts.

The treated accounts may happen to include unusually active users. Under complete randomization, for example, their pre-period mean might be 4.8 active days even though the full roster mean is 4.0. Randomization makes this discrepancy zero on average across assignments, but it is present in the experiment that actually occurred. If a prespecified relationship says that one additional pre-period active day predicts roughly two units of the experiment-period outcome, the 0.8-day imbalance suggests about 1.6 units of chance inflation in the treated outcome mean.

The same idea extends to a vector of pretreatment covariates. Define its HT estimate within the treated arm,

$$\widehat{\overline{X}}_{1,\mathrm{HT}} = \frac{1}{N}\sum_{i} \frac{Z_i X_i}{\pi_i}, \qquad \mathbb{E}_Z\Big(\widehat{\overline{X}}_{1,\mathrm{HT}}\Big) = \overline{X}.$$

The discrepancy $\overline{X} - \widehat{\overline{X}}_{1,\mathrm{HT}}$ is observable and has design expectation zero. If β converts covariate imbalance into predicted outcome imbalance, the natural correction is

$$\hat{\mu}_{1,\mathrm{GREG}}(\beta) = \hat{\mu}_{1,\mathrm{HT}} + (\overline{X} - \widehat{\overline{X}}_{1,\mathrm{HT}})^{\top}\beta. \tag{0.3}$$

Under complete randomization, $\widehat{\overline{X}}_{1,\mathrm{HT}}$ is simply the ordinary treated-arm covariate mean, so it is the 4.8 in the example. For a coefficient β fixed relative to assignment, Equation 0.3 is the **generalized-difference**, or regression-control, construction at the heart of GREG. Replacing β by a weighted regression estimate gives the usual **generalized regression estimator** (GREG) (Cassel et al. 1976). In the scalar example, $\beta = 2$ makes the correction $(4.0 - 4.8)2 = -1.6$.

Writing $m_1(x) = x^\top \beta$ gives an equivalent prediction-plus-residual form:

$$\hat{\mu}_{1,\mathrm{GREG}}(m_1) = \frac{1}{N}\sum_i m_1(X_i) + \frac{1}{N}\sum_i \frac{Z_i}{\pi_i}\{Y_i^{\mathrm{obs}} - m_1(X_i)\}. \tag{0.4}$$

The two terms divide the work. The full roster supplies the average prediction, which can be computed without observing any experiment-period outcomes. Inverse-probability weighting is then applied only to the prediction errors among treated units. If an intercept is desired, augment X_i with a leading 1. Under unequal probabilities, correcting imbalance in that intercept is exactly a correction for $\widehat{N}/N - 1$; under complete randomization, the intercept is already balanced. For β fixed before assignment,

$$\hat{\mu}_{1,\mathrm{GREG}}(\beta) - \overline{Y}(1) = \frac{1}{N}\sum_i \left(\frac{Z_i}{\pi_i} - 1\right)\{Y_i(1) - X_i^\top \beta\}.$$

Exact design unbiasedness therefore does not require the linear prediction to be correct: misspecification changes the residuals, but the IPW correction still centers their contribution correctly. Under independent Bernoulli assignment, the exact variance is

$$\mathrm{Var}_Z\{\hat{\mu}_{1,\mathrm{GREG}}(\beta)\} = \frac{1}{N^2}\sum_i \frac{1-\pi_i}{\pi_i}\{Y_i(1) - X_i^\top \beta\}^2.$$

HT is the case $\beta = 0$. At the other extreme, if the prediction were exact for every unit, every residual would be zero and the treated potential-outcome mean could be recovered from full-roster predictions without assignment noise. Between those extremes, the fixed-coefficient estimator reduces variance when the chosen coefficient leaves less design-weighted residual variation than the raw outcomes; an arbitrary or poorly transported coefficient need not help.

For control, replace Z_i/π_i by $(1-Z_i)/(1-\pi_i)$ and use a control-outcome prediction m_0. Subtracting the two prediction-plus-residual arm means estimates the ATE. This is the separate-arm version of the adjustment in Equation 0.2.

In practice, β is often fitted from the realized outcomes. Then the fixed-coefficient argument no longer establishes exact finite-sample unbiasedness. A coefficient fitted on historical data can be frozen before assignment; a coefficient fitted within the experiment needs the regression-adjustment qualifications discussed above. L03 uses the prediction-plus-IPW construction for both arms and names the resulting general form AIPW (Robins et al. 1994).

CUPED: choosing the control coefficient

The same online setting provides a natural scalar control variable: the outcome metric measured before the experiment. In this subsection, take X_i to be that single pre-period metric rather than a vector. **CUPED**, short for controlled-experiment using pre-experiment data, transforms the observed outcome to

$$Y_i^{\mathrm{cuped}}(\theta) = Y_i^{\mathrm{obs}} - \theta(X_i - \overline{X}).$$

Subtracting the same pretreatment quantity from $Y_i(1)$ and $Y_i(0)$ does not change unit i's treatment effect. Centering at $\overline{X}$ makes the transformed outcome retain the original finite-population mean within each treatment state. Taking the difference in arm means therefore gives, equivalently,

$$\hat{\tau}_{\mathrm{CUPED}}(\theta) = \hat{\tau}_{\mathrm{DM}} - \theta\hat{\tau}_X. \tag{0.5}$$

Here θ has the units of outcome per unit of the pre-period metric: it says how much outcome difference should be removed for one unit of chance covariate imbalance. For every θ fixed before assignment, this estimator remains design-unbiased because $\mathbb{E}_Z(\hat{\tau}_X) = 0$. The choice of θ controls precision. Expanding the variance of $\hat{\tau}_{\mathrm{DM}} - \theta\hat{\tau}_X$ gives

$$\begin{aligned}\mathrm{Var}_Z\{\hat{\tau}_{\mathrm{CUPED}}(\theta)\} = {} & \mathrm{Var}_Z(\hat{\tau}_{\mathrm{DM}}) + \theta^2\,\mathrm{Var}_Z(\hat{\tau}_X) \\ & - 2\theta\,\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X).\end{aligned} \tag{0.6}$$

When $\mathrm{Var}_Z(\hat{\tau}_X) > 0$, this quadratic is minimized at

$$\theta^* = \frac{\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X)}{\mathrm{Var}_Z(\hat{\tau}_X)}. \tag{0.7}$$

This is an oracle coefficient: its covariance with the treatment-effect estimator depends on the potential-outcome schedule and is not known before the experiment. At this value, the variance reduction relative to unadjusted difference in means is $\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X)^2 / \mathrm{Var}_Z(\hat{\tau}_X)$. Thus θ should be near zero when the pre-period measure does not predict experiment-period outcomes, and a poor coefficient can increase rather than decrease variance.

Under complete randomization, define the finite-population variance and arm-specific outcome–covariate covariances by

$$S_X^2 = \frac{1}{N-1}\sum_i (X_i - \overline{X})^2, \qquad S_{zX} = \frac{1}{N-1}\sum_i \{Y_i(z) - \overline{Y}(z)\}(X_i - \overline{X}).$$

The complete-randomization covariance identities are

$$\mathrm{Var}_Z(\hat{\tau}_X) = S_X^2\left(\frac{1}{N_1} + \frac{1}{N_0}\right),$$

and

$$\mathrm{Cov}_Z(\hat{\tau}_{\mathrm{DM}}, \hat{\tau}_X) = \frac{S_{1X}}{N_1} + \frac{S_{0X}}{N_0}.$$

The second expression has a plus sign because both $\hat{\tau}_{\mathrm{DM}}$ and $\hat{\tau}_X$ subtract their control-arm means. Substituting these identities into Equation 0.7 gives

$$\theta^* = \frac{S_{1X}/N_1 + S_{0X}/N_0}{S_X^2(1/N_1 + 1/N_0)} = \frac{N_0}{N}\frac{S_{1X}}{S_X^2} + \frac{N_1}{N}\frac{S_{0X}}{S_X^2}.$$

The oracle coefficient is therefore an allocation-weighted average of the two arm-specific slopes, with each slope weighted by the size of the opposite arm. Under equal allocation it is their simple average; if the two slopes coincide, their common value is optimal for any allocation. This calculation explains both why a pooled slope can work well and why treatment-specific slopes can improve on it.

The original CUPED recommendation is to use the same metric from a nearby pre-experiment period: it is often strongly correlated with the experiment-period metric while remaining unaffected by treatment (Deng et al. 2013). There are several practical ways to choose θ:

- **Estimate it before the experiment.** Use comparable historical experiments, a separate cohort, or relationships between earlier pretreatment windows, then freeze the coefficient before assignment. This preserves the exact fixed-coefficient design argument, but relies on stability across periods.
- **Estimate the oracle combination in the experiment.** Estimate a slope within each arm, say $\hat{\beta}_1$ and $\hat{\beta}_0$, and combine them as $(N_0/N)\hat{\beta}_1 + (N_1/N)\hat{\beta}_0$, mirroring the last expression for θ^*. In a balanced experiment this is the simple average. Regressing Y_i^{obs} on an intercept, Z_i, and $X_i - \overline{X}$ gives a familiar pooled common slope; including Z_i means that the slope is learned from within-arm outcome–covariate association rather than from the treatment/control mean difference. In a large balanced experiment it provides a convenient estimate of the same average when both arms inherit similar covariate dispersion; ordinary pooled regression need not reproduce the oracle weighting in an unbalanced experiment. Because the same outcomes choose the coefficient and estimate the effect, exact finite-sample unbiasedness is no longer automatic; inference should account for the fitted adjustment.
- **Allow treatment-specific slopes when warranted.** Include $Z_i(X_i - \overline{X})$, or fit the two arm-specific slopes directly. A single common θ can leave precision unused when the prognostic relationship differs by arm (Lin 2013).
- **Regularize or cross-fit richer adjustments.** With many baseline variables or a flexible predictor, assess predictive performance out of sample and use sample splitting or cross-fitting to limit overfitting. Cross-fitting does not by itself guarantee exact design unbiasedness, especially under fixed-total assignment. Only pretreatment variables belong in the adjustment.

Calibration: make the weighted arm reproduce its target

Return first to the completely randomized account experiment. Its treated-arm design weights are $d_i = 1/\pi_i = N/N_1$. They sum to N automatically, but if the treated users average 4.8 pre-period active days, their weighted covariate total is $N(4.8)$ rather than the known roster total $N(4.0)$. With unequal treatment probabilities, the same issue can affect both the total weight and the covariate total: design weights reproduce the full-roster targets in expectation over assignments, not necessarily in the realized assignment.

Calibration replaces the base weights by nearby weights that make selected weighted facts about the observed arm equal known full-roster targets. Its goal is not to make the treated arm resemble the roster in every respect. It enforces the particular comparisons encoded by the analyst's chosen features. In the scalar example, it chooses weights w_i satisfying

$$\sum_{i:Z_i=1} w_i = N, \qquad \sum_{i:Z_i=1} w_i X_i = N\overline{X}. \tag{0.1}$$

The first equation makes the weighted treated arm represent N accounts. The second makes its weighted pre-period total equal the known full-roster total. Under complete randomization the base weights already satisfy the first equation, but adjusting them to satisfy the second can disturb that total, so both constraints are imposed together.

For several covariates, first choose K scalar calibration features $h_1(X_i), \dots, h_K(X_i)$. These can be individual coordinates of X_i, indicators for categories, or prespecified transformations and interactions. They express which roster summaries the weighted arm should match. Collect them with an intercept into the $(K+1)$-dimensional column vector

$$g_i = \begin{pmatrix} 1 \\ h_1(X_i) \\ \vdots \\ h_K(X_i) \end{pmatrix}, \qquad t_g = \sum_{i=1}^{N} g_i.$$

The leading 1 is an intercept coordinate; g_i is an ordinary vector, not a binomial coefficient. Because the pretreatment covariates are known for the full experimental roster, the target total t_g is known. The calibration equation is the vector equality

$$\sum_{i:Z_i=1} w_i g_i = t_g. \tag{0.2}$$

Written component by component, it requires

$$\sum_{i:Z_i=1} w_i = N, \qquad \sum_{i:Z_i=1} w_i h_k(X_i) = \sum_{i=1}^{N} h_k(X_i), \quad k = 1, \ldots, K.$$

No binary-covariate assumption is involved. For a binary feature, the equation matches a count; for a continuous feature, it matches a weighted sum, and therefore its mean once the intercept constraint holds.

The constraints usually admit many weight vectors: matching $K+1$ totals does not by itself say which unit should absorb each adjustment. Let $\mathbf{w}$ collect the treated-arm weights. Calibration resolves that ambiguity by starting from the design weights d_i and moving them as little as possible while hitting the targets:

$$\min_{\mathbf{w}} \sum_{i:Z_i=1} d_i \, \phi\left(\frac{w_i}{d_i}\right) \quad \text{subject to} \quad \sum_{i:Z_i=1} w_i g_i = t_g, \tag{0.3}$$

where the convex function ϕ is minimized at 1. Thus the objective favors w_i/d_i near one, while the constraints force exactly as much movement as is needed for balance. With a strictly convex distance and feasible constraints, the solution is unique. Quadratic distance, $\phi(r) = (r-1)^2/2$, gives affine adjustments

$$w_i = d_i(1 + \eta^\top g_i),$$

whereas relative-entropy distance, $\phi(r) = r \log r - r + 1$, gives the positive exponential tilt

$$w_i = d_i \exp(\eta^\top g_i).$$

In each case $\eta \in \mathbb{R}^{K+1}$ is a vector of Lagrange-multiplier parameters chosen so that Equation 0.2 holds. Entropy calibration becomes classical raking when the features are indicators for categorical margins (Deville and Särndal 1992; Deville et al. 1993).

The resulting treated-mean estimator is

$$\hat{\mu}_{1,\text{cal}} = \frac{1}{N} \sum_{i:Z_i=1} w_i Y_i^{\text{obs}}.$$

Its potential advantage follows directly from the balance equations. This does not require assuming a linear outcome model. For any chosen β, define the residuals algebraically by

$$Y_i(1) = g_i^\top \beta + e_i.$$

Then

$$\hat{\mu}_{1,\text{cal}} - \overline{Y}(1) = \frac{1}{N} \left\{ \sum_{i:Z_i=1} w_i e_i - \sum_{i=1}^{N} e_i \right\}. \tag{0.4}$$

To obtain Equation 0.4, substitute $Y_i(1) = g_i^\top \beta + e_i$ into both the weighted observed total and the full-roster total. The two $g_i^\top \beta$ contributions are equal because of Equation 0.2, so they cancel. The only remaining error is imbalance in the residuals.

This identity gives the calibration objective its statistical meaning. If the chosen features predict outcomes well, their exact balance removes a systematic source of assignment noise. If $Y_i(1) = g_i^\top \beta$ exactly for every unit, all residuals are zero and the calibrated treated mean equals $\overline{Y}(1)$ exactly. With weakly predictive or overly numerous features, the benefit can be small or outweighed by unstable weights.

When the calibration problem is feasible and the relevant weighted feature matrix has full rank, quadratic distance gives the calibrated estimator an exact GREG representation with a corresponding weighted least-squares coefficient. Other smooth calibration distances generally produce different finite-sample estimators, although they can share a first-order GREG approximation under regularity conditions (Deville and Särndal 1992). Unlike the generalized-difference construction with a coefficient fixed before assignment, calibrated weights depend on the realized assignment, so exact design unbiasedness is not automatic.

The control arm can be calibrated separately to the same roster targets, beginning from $d_i = 1/(1-\pi_i)$. Subtracting the calibrated arm means estimates the ATE. More constraints are not always better: they can force extreme weights and inflate the residual variance. With nonnegative weights and an intercept constraint, the target covariate mean must lie in the observed arm's convex hull. Signed quadratic weights can reach targets outside that hull only when the target remains in the affine span of the observed feature vectors.

Diagnostics should therefore include residual balance, weight dispersion, and sensitivity to the feature set. For positive weights, one useful summary is the Kish effective sample size

$$N_{\text{eff}} = \frac{(\sum_{i:Z_i=1} w_i)^2}{\sum_{i:Z_i=1} w_i^2}.$$

It equals the observed arm size under equal weights and decreases as a small number of units carry more of the weighted total. It is a warning signal about weight concentration, not by itself a complete variance calculation.

What to carry forward

Many estimators in this lecture share one architecture, either exactly or through a first-order approximation:

$$\text{unbiased or identifying core} + \text{ a correction with known or approximately known mean zero.}$$

HT corrects unequal observation probabilities. Hájek stabilizes random total weight. Trotter–Tukey chooses how much to normalize. Regression, CUPED, GREG, and calibration remove predictable imbalance. Some corrections preserve exact unbiasedness only when their coefficients are fixed relative to assignment; fitted corrections generally require additional finite-sample or asymptotic analysis. Their differences matter, but seeing the common control-variate structure makes the menu navigable.

Advanced material

Practical experimentation: errors larger than the standard error

Large samples do not rescue a broken experiment. The practical program begins with Kohavi et al. (2007); Crook et al. (2009) then documents recurring failures in online controlled experiments: incorrect variance and power calculations, misleading percent-change intervals, bot traffic, instrumentation failures, ramp-up effects, and treatment/control populations that are not actually comparable. These are failures of measurement and implementation before they are failures of asymptotic approximation.

The replication evidence in Kohavi et al. (2026) adds a selection lesson. Eight large-scale tests of previously reported experimentation "patterns" had median samples in the millions and small minimum detectable effects, yet only two produced significant movement in the expected direction and one moved significantly in the opposite direction. The discrepancy is consistent with underpowered discovery studies, selective reporting, and the **winner's curse**: conditional on being selected for a large estimate, a noisy estimate is likely exaggerated.

Practical discipline follows: establish sample-ratio and instrumentation checks, specify the primary metric and analysis before looking, ramp cautiously, report uncertainty in interpretable units, and replicate claims that will guide many future experiments.

The propensity-score paradox

Why estimate an assignment probability that the experimenter already knows? The efficiency-bound analysis of Hahn (1998) sets up the issue. In a semiparametric superpopulation setting, Hirano et al. (2003) show that inverse-probability estimators using an estimated propensity score can be more efficient than otherwise analogous estimators using the true propensity.

There is no information paradox. Common likelihood and series estimators of the propensity score satisfy empirical fitting equations that can also impose useful covariate-balance conditions. Those conditions create a projection or control-variate correction, much like Equation 0.1 and Equation 0.2; arbitrary propensity estimation need not impose the same balance. The estimated propensity is not better knowledge of the randomization device. It is a route to an estimator that exploits realized covariate balance. In a randomized experiment, it is clearer to describe the gain directly as covariate adjustment or calibration, while retaining the known design probabilities as the basis for identification and inference.

A short history of inverse-probability normalization

Horvitz and Thompson used inverse inclusion probabilities to estimate finite-population totals exactly unbiasedly (Horvitz and Thompson 1952). Ratio estimation and Hájek normalization can trade finite-sample bias for stability; whether that trade is favorable depends on the design and finite population. Trotter and Tukey placed normalization on a continuum (Trotter and Tukey 1956), while Basu's survey-sampling arguments exposed how formally unbiased estimators can behave poorly when they ignore structural information (Basu 1971).

The same ratio appears in importance sampling: self-normalized importance sampling is stable and invariant to an unknown normalizing constant, but biased in finite samples (Owen 2013). Counterfactual learning rediscovered the same practical tradeoff when propensities are highly variable (Swaminathan and Joachims 2015). Adaptive normalization makes the amount of normalization estimable and connects its fixed point to regression control (Khan and Ugander 2023). The recurring problem is not domain-specific: random weights carry useful correction and avoidable noise at the same time.

Choosing estimators across families of experiments

Parikh et al. (2026) argue against choosing whichever estimator looks best inside a single realized trial. That practice turns estimator selection into another opportunity for overfitting. They instead use sample splitting across a **family** of randomized trials to compare estimators under a prespecified objective.

Their applications make the objective dependence concrete: weighted least squares performed best for inference-oriented criteria, while difference in means performed best for decision regret in the examined settings. The point is not a new universal winner. It is that "best estimator" is incomplete without a loss function, a relevant family of experiments, and a selection procedure separated from final evaluation.

Further reading

For the survey-sampling foundations, begin with Horvitz and Thompson (1952) and the exchange between Basu (1971) and Hájek (1971), then use Särndal et al. (1992) for the systematic development. The GREG and calibration line runs through Cassel et al. (1976), Deville and Särndal (1992), and Deville et al. (1993). For regression adjustment in experiments, reading Freedman (2008) together with Lin (2013) shows both the finite-sample concern and the modern interacted-regression resolution.

The normalization sequence from classical conditional Monte Carlo to adaptive IPW runs through Trotter and Tukey (1956) and Khan and Ugander (2023); Owen (2013) and Swaminathan and Joachims (2015) show the same self-normalization tradeoff in importance sampling and counterfactual learning. Deng et al. (2013) gives the original CUPED presentation in online experimentation.

For extensions beyond the main lecture path, read Hahn (1998) with Hirano et al. (2003) for the semiparametric efficiency logic behind estimated propensities. Kohavi et al. (2007), Crook et al. (2009), and Kohavi et al. (2026) connect estimator validity to experimentation practice and selective replication, while Parikh et al. (2026) treats estimator choice itself as a problem to be evaluated across a family of experiments.

Lecture brief: View the source `brief.md` · How these notes are made

References

L03: Experimental Design, Blocking, Balancing

Balance, rerandomization, and covariate-aware designs

Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- Use a grouped-assignment example to show why marginal treatment probabilities do not determine precision.
- Develop blocking and matched pairs as design-stage uses of pretreatment covariates.
- Contrast design-stage balance with regression and AIPW corrections made during analysis.
- Treat rerandomization as a complete design whose balance restrictions must also govern inference.

Ten coin flips disguised as one thousand

An experimental design is a probability distribution over complete assignment vectors, not merely a list of unit-level treatment probabilities. The distinction matters whenever assignments are dependent. As an extreme example, suppose $N = 1000$ units each have marginal treatment probability $1/2$. Partition them into ten groups of one hundred and flip one fair coin per group. Every unit in a group receives the same assignment.

The marginal probabilities match i.i.d. Bernoulli assignment, but the design contains only ten independent random decisions. Let $g = 1, \ldots, 10$ index the groups, let $\mathcal{J}_g$ be the set of units in group g, and let $A_g \in \{0, 1\}$ be the result of its coin flip. Thus $Z_i = A_g$ whenever $i \in \mathcal{J}_g$. With treatment probability $1/2$, the observable HT estimator (Horvitz and Thompson 1952) can be written first using observed outcomes and then, by consistency, using the corresponding potential outcomes:

$$\hat{\tau}_{\text{HT}} = \frac{1}{N}\sum_{i=1}^{N}\left\{2Z_i Y_i^{\text{obs}} - 2(1-Z_i)Y_i^{\text{obs}}\right\} = \frac{1}{N}\sum_{i=1}^{N}\left\{2Z_i Y_i(1) - 2(1-Z_i)Y_i(0)\right\}.$$

The second expression may look as though it uses both potential outcomes, but its assignment indicators select only the observed one: the $Y_i(0)$ term vanishes for a treated unit and the $Y_i(1)$ term vanishes for a control unit. Subtracting the fixed estimand and collecting units that share a coin gives the exact error representation

$$\hat{\tau}_{\text{HT}} - \tau = \frac{1}{N}\sum_{g=1}^{10}(2A_g - 1)\sum_{i\in\mathcal{J}_g}\{Y_i(1) + Y_i(0)\}.$$

The inner sum is a fixed group total; all assignment randomness in that group is the single sign $2A_g - 1$. Independence of the ten signs therefore gives

$$\text{Var}_Z(\hat{\tau}_{\text{HT}}) = \frac{1}{N^2}\sum_{g=1}^{10}\left[\sum_{i\in\mathcal{J}_g}\{Y_i(1) + Y_i(0)\}\right]^2. \tag{0.1}$$

Under i.i.d. unit-level Bernoulli assignment, the corresponding variance is

$$\text{Var}_Z(\hat{\tau}_{\text{HT}}) = \frac{1}{N^2}\sum_{i=1}^{N}\{Y_i(1) + Y_i(0)\}^2. \tag{0.2}$$

The contrast between Equation 0.1 and Equation 0.2 is the location of the square. In the grouped design, outcomes are summed within a group and then squared, so the one hundred units sharing a coin move together. Under unit-level assignment, each unit contributes its own squared term.

Take the stark special case $Y_i(1) = a$ and $Y_i(0) = b$ for every unit. The grouped design has variance $(a+b)^2/10$. That is exactly the i.i.d. Bernoulli variance for **ten** units, not one thousand. I.i.d. Bernoulli assignment of all one thousand units has variance $(a+b)^2/1000$.

This exact equivalence is specific to HT in the example with constant potential outcomes. It partly reflects HT's random total weight: conditional on at least one treated and one control group, difference in means is exactly $a-b$ and has zero variance. With heterogeneous group totals, however, every estimator's assignment variation still originates in only ten coin flips. The example therefore combines two lessons from L02 and L03: normalization matters, and dependence determines the effective source of randomization.

The point is not that dependence is always harmful. The positive dependence in this example makes many units move together. Negative dependence, by contrast, can force treatment and control counts or covariate totals to balance and thereby improve precision. What determines these properties is the **joint distribution of the whole assignment vector Z**, not the marginal probabilities considered one unit at a time.

Blocking: randomize among comparable units

The grouped example used dependence badly: it made every unit in a group receive the same treatment. Blocking uses dependence for the opposite purpose. Imagine a student experiment in which prior achievement is strongly predictive of the outcome. A simple complete randomization can, by chance, place more high-achieving students in one arm. Before assigning treatment, we can divide students into groups with similar prior achievement and require both treatment arms to appear within every group. The comparison is then made among students who were similar before treatment.

Formally, suppose pretreatment covariates divide the finite population into H nonoverlapping **blocks**, also called strata. Let $\mathcal{I}_h$ denote the units in block h and $N_h = |\mathcal{I}_h|$ its size. We choose in advance to treat exactly N_{1h} units and control $N_{0h} = N_h - N_{1h}$ units in that block. **Stratified complete randomization**, or blocking, performs complete randomization independently within each $\mathcal{I}_h$. Thus units in different blocks are randomized separately, while assignments inside a block are negatively dependent because their treated count is fixed.

For arm $z \in \{0, 1\}$, define the observed mean within block h by

$$\overline{Y}_{zh}^{\text{obs}} = \frac{1}{N_{zh}} \sum_{i \in \mathcal{I}_h} \mathbf{1}\{Z_i = z\} Y_i^{\text{obs}},$$

where $\mathbf{1}\{Z_i = z\}$ is the indicator that unit i was assigned to arm z, and N_{zh} is the fixed number assigned to that arm in block h. The within-block treatment-control contrast is

$$\hat{\tau}_h = \overline{Y}_{1h}^{\text{obs}} - \overline{Y}_{0h}^{\text{obs}}$$

Define the block-specific finite-population effect

$$\tau_h = \frac{1}{N_h} \sum_{i \in \mathcal{I}_h} \{Y_i(1) - Y_i(0)\}.$$

Complete randomization within the block makes $\hat{\tau}_h$ design-unbiased for τ_h. The population ATE is not an equally weighted average of the block effects unless all blocks have the same size: every unit, rather than every block, must receive equal weight. This yields

$$\hat{\tau}_{\text{block}} = \sum_{h=1}^{H} \frac{N_h}{N} \hat{\tau}_h. \tag{0.1}$$

The weights N_h/N are the fractions of experimental units in the blocks, so $\tau = \sum_h (N_h/N)\tau_h$ and Equation 0.1 is design-unbiased for τ. To describe its precision, let S_{zh}^2 be the finite-population

variance of $Y_i(z)$ among units in block h, and let $S^2_{\tau h}$ be the within-block variance of the individual effects τ_i. Applying Neyman's finite-population variance calculation inside each block and then adding across the independently randomized blocks gives (Splawa-Neyman et al. 1990)

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{block}}) = \sum_{h=1}^{H} \left(\frac{N_h}{N}\right)^2 \left(\frac{S^2_{1h}}{N_{1h}} + \frac{S^2_{0h}}{N_{0h}} - \frac{S^2_{\tau h}}{N_h}\right). \tag{0.2}$$

The factor $(N_h/N)^2$ appears because each random within-block contrast enters the overall estimator multiplied by N_h/N. As in L02, $S^2_{\tau h}$ is generally unidentified: observing only one potential outcome per unit does not reveal the within-block variance of individual treatment effects. A usable conservative estimator drops these nonnegative terms and substitutes the observed within-block sample variances s^2_{zh} for S^2_{zh}:

$$\hat{V}_{\mathrm{Neyman}} = \sum_{h=1}^{H} \left(\frac{N_h}{N}\right)^2 \left(\frac{s^2_{1h}}{N_{1h}} + \frac{s^2_{0h}}{N_{0h}}\right).$$

This plug-in estimator requires enough observed units to estimate both within-block arm variances. In a matched-pair design there is only one observed outcome per arm in each pair, so uncertainty must instead be estimated from the collection of randomized pair contributions.

Blocking helps when the blocks capture pretreatment differences that predict outcomes. For example, if outcomes contain a large block-specific baseline component, that component is shared by the treated and control units compared within the block and cancels from their contrast. Between-block baseline differences therefore do not inflate the treatment comparison. Blocking also fixes arm counts within every block. With a common treatment fraction, it exactly balances the categorical block indicators; continuous covariates used to construct the blocks are balanced only approximately, to the extent that units placed together are similar.

Blocking is not free. Many fine blocks can be operationally difficult; attrition can break their intended structure; and a block must contain enough units to support the planned arm counts. Blocking on weakly prognostic variables can provide little gain. The design should also be preserved in the analysis: estimate effects within blocks and aggregate using the target-population weights.

Matched pairs

Matched pairs are the limiting case $N_h = 2$ and $N_{1h} = 1$. Suppose there are $H = N/2$ pairs, and label the two units in pair h by a and b. Let $A_h = 1$ when unit a is treated and $A_h = 0$ when unit b is treated. The signed observed difference

$$D_h = (2A_h - 1)\,(Y^{\mathrm{obs}}_{ha} - Y^{\mathrm{obs}}_{hb})$$

always means "treated outcome minus control outcome": when $A_h = 1$, it is $Y^{\mathrm{obs}}_{ha} - Y^{\mathrm{obs}}_{hb}$; when $A_h = 0$, the sign reverses. Its design expectation is the average of the two unit-level effects in pair h. Consequently,

$$\hat{\tau}_{\text{pair}} = \frac{1}{H} \sum_{h=1}^{H} D_h$$

is design-unbiased for the finite-population ATE. Because each pair supplies only one treated and one control observation, there is no within-arm variance estimate inside a pair. Precision is instead learned from how the randomized contributions D_h vary across pairs.

Pairing can be powerful because assignment only decides which of two deliberately similar units is treated. If the matched units have similar potential outcomes under each arm, the two possible values of D_h are close, reducing its assignment variance. Pairing concentrates randomization locally: each pair has only two possible assignments, although assignments remain independent across pairs. Bai (2022) characterizes precision-optimal matched-pair designs, while Bai et al. (2022) develops inference that respects the pairing. Poor matches can lock in bad comparisons, and missing outcomes are especially damaging. Matching should therefore be based on pretreatment information chosen for prognostic value, not on an urge to balance every measured variable.

Design balance versus analysis adjustment

Return to the prior-achievement example. We can use the baseline score **before assignment**, by blocking students with similar scores, or **after assignment**, by statistically correcting whatever baseline-score difference happened to arise. These are different interventions on the experiment. The first changes the assignment distribution; the second leaves the design alone and changes the estimator. The broader design-first principle is articulated by Rubin (2008), while Bruhn and McKenzie (2009) compares common randomization procedures in field-experiment practice:

- **At design:** restrict or structure assignments so covariate imbalance is small by construction.
- **At analysis:** allow a simpler assignment design and subtract the part of the outcome difference predicted by realized covariate imbalance.

To make the analysis-stage option precise, suppose X_i is a column vector of pretreatment covariates, $\overline{X}_z^{\text{obs}}$ is its observed sample mean in arm z, and β is a coefficient vector of the same dimension. Under two-arm complete randomization, consider the regression-adjusted form

$$\hat{\tau}_{\text{adj}}(\beta) = \hat{\tau}_{\text{DM}} - \beta^{\top}(\overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}).$$

The vector $\overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}}$ is the realized covariate imbalance. Its inner product with β is the part of the raw outcome difference attributed to that imbalance, so the estimator subtracts it. If β is fixed with respect to assignment, the correction has design expectation zero because complete randomization gives each arm the same expected covariate mean. A coefficient aligned with the outcome–covariate relationship can therefore reduce assignment variance without changing the design expectation; a poorly chosen coefficient can instead increase variance.

A coefficient fitted from the experiment's outcomes is assignment-dependent and needs the finite-sample or asymptotic justification discussed in L02 (Freedman 2008; Lin 2013). Design balance is visible before outcomes are observed and protects a range of later analyses. Analysis adjustment can use continuous, high-dimensional, or nonlinear predictions more flexibly, but it requires appropriate estimation and uncertainty calculations. The two are complements: one can block on a few decisive variables and adjust for remaining prognostic variation.

This comparison also reveals a general principle. Design validity does not require a correct outcome model relating Y to X; its precision gains do. Balancing weak or wrongly represented features may provide little benefit, while concentrating the design too aggressively around one criterion can reduce robustness to outcome structure that criterion missed.

AIPW: prediction plus randomized correction

Covariate adjustment presents a useful tension. An outcome model can predict a missing potential outcome for every unit, but a misspecified model need not average to the correct finite-population mean. An HT estimator has the opposite strength: known assignment probabilities make it design-unbiased, but it may be noisy because it inverse-weights the full observed outcomes. **Augmented inverse probability weighting** (AIPW) combines these two starting points. It uses predictions to remove explainable variation and uses randomization to correct the predictions rather than trusting them outright.

The target for arm z is the finite-population mean $\overline{Y}(z)$. Let $m_z(x)$ predict the potential outcome under arm z from pretreatment covariates. For the moment, treat the N numbers $m_z(X_i)$ as fixed before assignment; they need not equal the actual potential outcomes. Also assume the relevant positivity condition

$$\pi_i(z) = \mathbb{P}_Z(Z_i = z) > 0 \qquad \text{for every unit and arm under study.}$$

Here $\pi_i(z)$ is the known probability that the design assigns unit i to arm z. Positivity ensures that every unit whose potential outcome contributes to $\overline{Y}(z)$ has some chance of revealing that outcome.

The population average of the predictions, $N^{-1}\sum_i m_z(X_i)$, is available because the covariates are observed for all experimental units. Correct it using the prediction errors observed in arm z:

$$\hat{\mu}_{z,\text{AIPW}} = \frac{1}{N}\sum_{i=1}^{N}\left[m_z(X_i) + \frac{\mathbf{1}\{Z_i = z\}}{\pi_i(z)}\{Y_i^{\text{obs}} - m_z(X_i)\}\right], \tag{0.1}$$

The indicator $\mathbf{1}\{Z_i = z\}$ is one exactly when unit i is assigned to arm z. Thus the second term is zero for units outside that arm and equals the observed residual divided by its assignment probability for units inside it. Subtracting the two arm-specific mean estimators gives

$$\hat{\tau}_{\text{AIPW}} = \hat{\mu}_{1,\text{AIPW}} - \hat{\mu}_{0,\text{AIPW}}$$

as an estimator of $\tau = \overline{Y}(1) - \overline{Y}(0)$ (Robins et al. 1994; Tsiatis et al. 2008). The GREG construction in L02 has the same prediction-plus-weighted-residual architecture; AIPW writes it arm by arm and allows general prediction functions.

The construction can be read in three steps:

1. predict every unit's potential outcome under arm z;
2. among units actually observed in arm z, inverse-weight the prediction error;
3. add that correction to the population average prediction.

The correction works because, for each unit separately, inverse-probability weighting recovers the full prediction error. Since $Y_i^{\text{obs}} = Y_i(z)$ on the event $Z_i = z$,

$$\mathbb{E}_Z\left[\frac{\mathbf{1}\{Z_i = z\}}{\pi_i(z)}\{Y_i^{\text{obs}} - m_z(X_i)\}\right] = Y_i(z) - m_z(X_i).$$

Adding the fixed prediction back therefore gives design unbiasedness without assuming that the prediction is correct:

$$\begin{aligned}\mathbb{E}_Z(\hat{\mu}_{z,\text{AIPW}}) &= \frac{1}{N}\sum_i [m_z(X_i) + Y_i(z) - m_z(X_i)] \\ &= \overline{Y}(z).\end{aligned}$$

This is the central division of labor. The model supplies a precision device; the randomization correction supplies validity for the finite-population target. When $m_z(X_i)$ is close to $Y_i(z)$, the residuals passed through the random inverse-probability weights are smaller and the estimator can be more precise than HT. A poor fixed predictor does not destroy design unbiasedness, though it can increase variance.

> ⚠ Fitted predictions are part of the estimator
>
> The exact argument above treats m_z as fixed with respect to assignment. An externally trained or otherwise preassignment predictor preserves that exact conditional design-unbiasedness. If the same randomized outcomes are used to fit m_z, exact finite-sample unbiasedness does not follow automatically. Under independent assignment, carefully constructed leave-one-out or cross-fitted predictions can separate a unit's assignment from its prediction. Under complete, blocked, or rerandomized designs, assignments are dependent, so cross-fitting alone is not a universal finite-sample fix; one needs a design-specific argument or suitable asymptotics. Randomization remains the foundation, but "the model may be wrong" does not mean "the fitting procedure is irrelevant."

In observational causal inference, AIPW is often introduced through double robustness. That is not the focus here: the assignment probabilities in a randomized experiment are known. Our emphasis is the control-variate interpretation—prediction for precision, inverse-probability correction for the design-based target.

Pair matching in cluster-randomized experiments

Sometimes treatment is assigned to schools, villages, or clinics while outcomes are measured on individuals. With few clusters, ordinary cluster randomization can leave large baseline differences. Imai et al. (2009) argue for pairing similar clusters and randomizing one cluster within each pair; Middleton and Aronow (2015) develops the design-based estimand and weighting issues for cluster-randomized experiments.

The distinction between the unit of assignment and the unit of measurement is essential. Suppose there are K pairs indexed by $k = 1, \ldots, K$, and pair k contains clusters $j = 1, 2$. Cluster (k, j) contains n_{kj} individuals, and $N = \sum_{k=1}^{K} \sum_{j=1}^{2} n_{kj}$ is the total number of individuals. Under cluster-level treatment and no interference between clusters, define the cluster potential-outcome total

$$T_{kj}(z) = \sum_{\ell=1}^{n_{kj}} Y_{kj\ell}(z),$$

where ℓ indexes individuals within a cluster. The individual-weighted finite-population ATE is then

$$\tau = \frac{1}{N} \sum_{k=1}^{K} \sum_{j=1}^{2} \{T_{kj}(1) - T_{kj}(0)\}.$$

Let $A_k = 1$ mean that cluster 1 in pair k is treated; when $A_k = 0$, cluster 2 is treated. The two possibilities occur with probability $1/2$, independently across pairs. Write T_{kj}^{obs} for the total under the assignment cluster (k, j) actually receives. The HT estimator of the individual-weighted ATE is

$$\hat{\tau}_{\text{HT}} = \frac{2}{N} \sum_{k=1}^{K} (2A_k - 1)\{T_{k1}^{\text{obs}} - T_{k2}^{\text{obs}}\}.$$

The sign $2A_k - 1$ always orients the expression as treated-cluster total minus control-cluster total. The leading factor 2 is the inverse of each cluster's treatment or control probability. Averaging over the two assignments in pair k recovers the sum of the treatment effects for all individuals in both clusters; dividing by N therefore targets the individual-weighted ATE above.

Each pair contributes one independent randomized contrast, so design-based uncertainty is built from variation across those pair contributions. Two details are easy to miss:

- The effective randomization units are clusters, not the individuals nested within them.
- Weighting determines the estimand. The total-based HT expression targets an average over individuals. Averaging cluster means equally instead targets a cluster-weighted or pair-weighted effect; unequal cluster sizes make the distinction consequential.

Pair matching can substantially improve precision when matches are prognostic, but the analysis must honor both pairing and unequal cluster sizes. Treating all individual observations as independently randomized produces fictitious precision.

Rerandomization

Blocking is attractive when units fall naturally into small groups within which fixed treatment counts can be enforced. With several continuous covariates, however, useful blocks may be difficult to define. **Rerandomization** takes a different route: repeatedly draw an assignment from a familiar proposal design, measure its global covariate imbalance, and retain the draw only if that imbalance meets a prespecified standard (Morgan and Rubin 2012). It changes the design before outcomes are seen; it is not post hoc deletion of an inconvenient randomization.

Write $\mathbb{P}_{\mathrm{CR}}$ for the proposal complete-randomization design and assume equal-sized treatment arms. For the pretreatment covariate vector X_i, let

$$\hat{\tau}_X = \overline{X}_1^{\mathrm{obs}} - \overline{X}_0^{\mathrm{obs}}$$

be the vector of treated-minus-control covariate mean differences. Zero is perfect mean balance. Different coordinates may have different units and may be strongly correlated, so an ordinary Euclidean norm is not a meaningful common scale. Assuming the covariance matrix is nonsingular, the Mahalanobis balance score (Mahalanobis 1936) standardizes the imbalance by its covariance under the proposal design:

$$M(\mathbf{Z}) = \hat{\tau}_X^{\top} \left\{ \mathrm{Cov}_{\mathrm{CR}}(\hat{\tau}_X) \right\}^{-1} \hat{\tau}_X. \tag{0.1}$$

The covariance subscript "CR" matters: $\mathrm{Cov}_{\mathrm{CR}}(\hat{\tau}_X)$ describes how covariate imbalance would vary over repeated proposal assignments, with the covariates held fixed. Consequently, $M(\mathbf{Z})$ is a scalar standardized squared distance from exact balance. With a single covariate, it is simply the squared mean difference divided by its complete-randomization variance.

Choose a threshold a before assignment so that the acceptance event has positive proposal probability. Repeatedly draw a complete randomization and accept the first draw satisfying $M(\mathbf{Z}) \le a$. The actual rerandomization design, denoted $\mathbb{P}_{\mathrm{ReM}}$, is the proposal distribution conditional on acceptance:

$$\begin{aligned} \mathbb{P}_{\mathrm{ReM}}(\mathbf{Z} = \mathbf{z}) &= \mathbb{P}_{\mathrm{CR}}\{\mathbf{Z} = \mathbf{z} \mid M(\mathbf{Z}) \le a\} \\ &= \frac{\mathbb{P}_{\mathrm{CR}}(\mathbf{Z} = \mathbf{z}) \mathbf{1}\{M(\mathbf{z}) \le a\}}{\mathbb{P}_{\mathrm{CR}}\{M(\mathbf{Z}) \le a\}}. \end{aligned} \tag{0.2}$$

If complete randomization is uniform over assignments with the prescribed arm sizes, rerandomization is uniform over the subset that passes the balance rule. This is still a randomized design, provided the covariates, score, and threshold are fixed in advance and the acceptance procedure is recorded. Ad hoc rejection until the groups merely "look nice" does not define a reproducible assignment law.

What rerandomization buys

Let $\mathbf{1}_N$ denote the length-N vector of ones, so the treatment-control complement of $\mathbf{z}$ is $\mathbf{1}_N - \mathbf{z}$. If the acceptance rule is symmetric—$\mathbf{z}$ is accepted exactly when its complement is—then every accepted assignment has an equally likely accepted partner that reverses all arm labels. With equal arm sizes, each unit therefore retains marginal treatment probability 1/2, and difference in means remains design-unbiased. The joint assignment distribution has nevertheless changed: assignments with large $M(\mathbf{Z})$ now have probability zero. When the balanced covariates predict outcomes, removing those assignments can substantially reduce variance.

But the analysis must use the design that was actually run. An exact Fisher randomization test of a sharp null should sample from accepted assignments, not from all complete randomizations. Under the regularity conditions studied by Li et al. (2018), the usual Neyman normal interval calculated as if complete randomization had been used is asymptotically conservative; this is not a universal finite-sample statement.

The large-sample result of Li et al. (2018) explains the mechanism. Decompose the outcome variation into a component linearly associated with the balanced covariates and a residual component orthogonal to them. Under their regularity conditions, the difference-in-means fluctuation likewise separates into two independent pieces. The residual piece remains asymptotically Gaussian because the acceptance rule does not directly constrain it. The covariate-associated piece has a Gaussian limit truncated to the accepted balance region. Rerandomization therefore shrinks precisely the part of the estimator's variation aligned with X. The precision gain is larger when that part explains more of the outcome and when the threshold is more stringent. Li and Ding (2020) then studies how regression adjustment and rerandomization interact rather than treating design- and analysis-stage balance as substitutes.

The robustness tradeoff

Rerandomization has an appealing operational story and preserves multiple acceptable assignments. Its weaknesses are equally instructive:

- a stringent threshold lowers the acceptance probability and concentrates the assignment distribution;
- Mahalanobis balance is only as relevant as the chosen covariates and feature representation; in high dimensions, covariance regularization also changes the metric;
- for a fixed acceptance probability, increasing dimension weakens the attainable improvement in each direction and makes covariance estimation or regularization more consequential;
- unrecorded changes to the acceptance rule invalidate the stated design;
- analysis that pretends complete randomization was used mischaracterizes uncertainty.

The threshold therefore mediates balance against robustness. Making a smaller improves balance along the chosen covariate directions, but it leaves fewer possible assignments and makes the design more dependent on whether those directions capture the outcome structure that matters. Here robustness includes both protection against a misspecified balance criterion and enough randomness to support credible, reproducible inference. Perfect observed balance is not the only goal of a randomized experiment.

What to carry forward

Experimental design controls the dependence among assignments. Blocking uses scientifically meaningful local comparisons; matching makes those blocks as small as possible; rerandomization conditions on acceptable global balance. Analysis adjustment and AIPW can recover additional precision after a simpler design. In every case, the estimator and its uncertainty calculation must correspond to the assignment distribution that was actually used.

Advanced material

Gram–Schmidt walk designs

The word **discrepancy** describes a signed imbalance. To see the connection to experiments, suppose the treatment and control arms have equal size and define the assignment sign $s_i = 2Z_i - 1$, which is $+1$ for treatment and -1 for control. Then

$$\hat{\tau}_X = \overline{X}_1^{\text{obs}} - \overline{X}_0^{\text{obs}} = \frac{2}{N} \sum_{i=1}^{N} s_i X_i.$$

Thus choosing signs that make the vector sum $\sum_i s_i X_i$ small is exactly choosing an assignment with small covariate mean imbalance. Discrepancy theory studies how signs can be chosen so that many coordinates or other linear summaries remain small simultaneously. The norm used to judge the signed sum specifies which kinds of imbalance matter.

A purely discrepancy-minimizing algorithm might return one deterministic assignment. The Gram–Schmidt walk (GSW) instead turns a discrepancy-control procedure into a randomized assignment design (Harshaw et al. 2024). It generates the signs through a random walk while steering their covariate sum toward balance.

The design has a tuning parameter that governs how aggressively the walk prioritizes covariate balance relative to unconstrained randomization. At one extreme it behaves more like conventional random assignment; at the other it pushes harder to keep the linear discrepancy small. Crucially, it retains prescribed unit-level marginals, such as $\mathbb{P}_Z(Z_i = 1) = 1/2$, while inducing dependence among assignments. The resulting guarantees concern design-based mean squared error over specified classes of outcome structure. This separates GSW from selecting a single best-looking assignment: its balance comes from a probability distribution over assignments, not from abandoning randomization.

The guarantee also states its own limitation: linear balance protects outcomes whose conditional mean is well represented by the chosen linear features. Cytrynbaum (2026) shows that uniformly approaching classical efficiency over broad smooth outcome classes becomes impossible once dimension grows too quickly. The paper develops a nonparametric, kernelized GSW in a reproducing-kernel Hilbert space and identifies more favorable rates under additive structure. Kernelization expands the function class being balanced; it does not eliminate the need to say which structure is plausible.

Correlated cluster designs without rich covariates

The paired-cluster discussion used covariates to decide which clusters should be compared. Candogan et al. (2024) ask a different question: can dependence be designed robustly when cluster size is the principal available structure? They study correlated cluster assignments with equal marginal treatment probabilities and a cluster-aware HT estimator, choosing dependence to minimize a worst-case variance criterion.

Their imbalance-reducing construction uses cluster sizes rather than a rich outcome-prediction model. The asymptotic guarantees require regularity, including that no single cluster dominate the population. The result is therefore neither covariate matching nor a claim that size is always the most prognostic variable. It shows that assignment correlation can be optimized against a stated worst-case class even when conventional matching information is sparse.

Optimization-based designs and the value of randomness

Once discrepancy has been written as a function of the assignment signs, one can search directly for the assignment minimizing it—for example, minimizing a norm of $\sum_i s_i X_i$ subject to fixed arm sizes. Mixed-integer optimization can produce striking balance in small samples (Bertsimas et al. 2015), and minimax formulations connect classes of plausible outcome functions to the imbalance metric that should be optimized (Kallus 2018). The objective is never neutral: choosing a norm, feature map, or function class says which outcome patterns the design is intended to protect.

Selecting one minimizing assignment eliminates design-based randomization. Randomizing between that assignment and its treatment-control complement restores unit-level marginals of 1/2, but leaves only two assignments in the support. Either construction can be brittle to aspects of the outcome surface omitted from the objective. More diffuse randomization distributes that risk. Kallus (2021) formalizes this through mixed-strategy optimal designs: with no informative structure, complete randomization is minimax; gains require restrictions on plausible outcome functions, and inference supplies a reason to retain sufficiently diffuse randomization.

GSW, rerandomization, and mixed-strategy optimization occupy different points on the same frontier. Each asks how much imbalance to remove while preserving enough randomness for robustness and inference.

Efron's biased coin

Efron's biased-coin design addresses a sequential problem: how can an experiment resist persistent arm-size imbalance without making its next assignment completely predictable? (Efron 1971) After n assignments, let $D_n = N_{1n} - N_{0n}$ be the treated count minus the control count. If $D_n = 0$, assign the next unit by a fair coin. If $D_n > 0$, treatment is overrepresented, so assign the next unit to control with a chosen probability $q > 1/2$; if $D_n < 0$, assign it to treatment with that probability. The parameter q controls the pull back toward equal counts.

At $q = 1/2$, the feedback disappears and the rule is ordinary i.i.d. Bernoulli assignment. As q rises toward one, balance is restored more forcefully but future assignments become easier to predict. The design therefore makes the balance–randomness tradeoff explicit. Covariate-adaptive procedures

such as Pocock and Simon (1975) generalize the same feedback idea to prognostic-factor imbalance, but the dependence created by that feedback must be reflected in the analysis.

D-, A-, and E-optimal design

Section 7.5 of Boyd and Vandenberghe (2004) gives a modern convex-optimization treatment of a classical model-based design problem whose general theory traces to Kiefer (1959). Suppose measurement option r has a known regressor vector v_r, and let $\lambda_r \geq 0$ be the fraction of measurement effort allocated to that option, with $\sum_r \lambda_r = 1$. In a homoskedastic linear model, the normalized information matrix is

$$\mathcal{I}(\lambda) = \sum_r \lambda_r v_r v_r^\top.$$

When this matrix is nonsingular, the coefficient covariance is proportional to $\mathcal{I}(\lambda)^{-1}$. The design problem chooses the allocation λ to make that covariance small, but a matrix is not ordered by a single universally best scalar. Common criteria emphasize different geometries:

- **D-optimality** maximizes $\det\{\mathcal{I}(\lambda)\}$, equivalently shrinking the volume of the coefficient confidence ellipsoid.
- **A-optimality** minimizes $\mathrm{tr}\{\mathcal{I}(\lambda)^{-1}\}$, the sum of the marginal coefficient-variance proxies.
- **E-optimality** maximizes the smallest eigenvalue of $\mathcal{I}(\lambda)$, protecting the least-informed coefficient direction.

These criteria are connected to covariate-balanced treatment design through convex optimization and worst-direction reasoning. They are not the same problem. Classical optimal design begins from a response model and chooses where to measure parameters; GSW chooses a randomized treatment allocation and seeks design-based protection over outcome classes. The analogy is most useful when it makes the assumed geometry explicit.

Staggered rollouts beyond SUTVA

In a staggered rollout, clusters or units adopt treatment at different assigned times. The stepped-wedge formulation of Hussey and Hughes (2007) is a canonical example. The design creates comparisons across adoption cohorts and calendar periods. Operational constraints often motivate it, but the schedule can also generate deliberate variation in treatment saturation and exposure duration.

That variation is valuable when no-interference or no-carryover assumptions are implausible: outcomes can be indexed by neighbors' adoption, local saturation, or time since treatment rather than by a single binary label. The cost is that calendar-time trends, anticipation, and dynamic effects become part of the estimand and design. A rollout is therefore not merely a delayed two-arm experiment; it requires a potential-outcome table indexed by assignment histories.

Further reading

For historical orientation, Fisher (1935) develops randomization and blocking, while Rubin (2008) states the modern design-before-analysis principle. Ding (2023) provides a concise contemporary treatment of finite-population experiments and rerandomization. For matched designs, read Bai (2022) and Bai et al. (2022) alongside the cluster-randomized perspective of Imai et al. (2009) and Middleton and Aronow (2015).

For rerandomization, begin with Morgan and Rubin (2012) for the design principle, then read Li et al. (2018) for the non-Gaussian asymptotic distribution induced by acceptance and Li and Ding (2020) for its interaction with regression adjustment. Candogan et al. (2024) shows how cluster-assignment dependence can instead be optimized from cluster-size information and a worst-case criterion.

The modern balance–robustness sequence runs from optimization-based designs (Bertsimas et al. 2015; Kallus 2018, 2021) to randomized discrepancy control (Harshaw et al. 2024) and its recent nonlinear extension (Cytrynbaum 2026). Efron (1971) and Pocock and Simon (1975) provide sequential counterparts: use randomization feedback to resist arm-size or covariate imbalance without becoming deterministic. For classical information-matrix optimality, pair Kiefer (1959) with the computational treatment in Boyd and Vandenberghe (2004); for staggered cluster rollouts, see Hussey and Hughes (2007).

Lecture brief: View the source `brief.md` · How these notes are made

References

L04: Positivity, Trimming, Target Populations, and Partial Identification

Overlap, target populations, and bounds

Lecture brief: View the source `brief.md` · How these notes are made

> **Note**
>
> Lecture notes for this session have not yet been generated.

L05: Interference

Exposure mappings, testing, and experimental design

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> **Note**
>
> Lecture notes for this session have not yet been generated.

L06: Estimation with Structural Models

Low-order structure, monotonicity, and submodularity under interference

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> Lecture notes for this session have not yet been generated.

L07: Policy Evaluation and Panel Data

Treatment rules, heavy-tailed outcomes, and panel estimators

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> **Note**
>
> Lecture notes for this session have not yet been generated.

L08: Advanced Estimation Topics

Doubly robust estimation, orthogonalization, and covariate weighting

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>
> Lecture notes for this session have not yet been generated.

L09: AI for Causal Inference

A structured survey of a rapidly changing literature

Lecture brief: View the source `brief.md` · How these notes are made

> **Note**
>
> Lecture notes will be generated after a current literature search.

Notation

The conventions below are maintained in the course-wide `NOTATION.md`. They are extended as later lectures establish new notation.

Course notation

This file records notation intended to remain stable across lectures. Before introducing notation in a lecture, check the conventions here. Add notation when it becomes course-wide; leave symbols used only in a local derivation or example in that lecture.

Units, samples, and populations

- N is the number of units in the finite experimental population, indexed by $i = 1, \dots, N$.
- X_i is the vector of pretreatment covariates for unit i; $\mathbf{X}$ collects these vectors.
- A target population beyond the experimental units is described explicitly. Do not use "ATE" without making its target population clear.

Treatments, assignments, and exposures

- $Z_i \in \{0, 1\}$ is unit i's treatment assignment, with 1 denoting treatment and 0 control. The assignment vector is $\mathbf{Z} = (Z_1, \dots, Z_N)$.
- $N_1 = \sum_{i=1}^{N} Z_i$ and $N_0 = N - N_1$ are the realized treatment-arm sizes.
- $\mathcal{Z}$ is the support of the experimental design, the set of assignment vectors with positive probability.
- When units are partitioned into H design strata or blocks, $h = 1, \dots, H$ indexes the strata, N_h is the stratum size, and N_{1h} and N_{0h} are its treatment-arm sizes.

Potential outcomes and observed outcomes

- $Y_i(z)$ is the potential outcome for unit i under treatment level $z \in \{0, 1\}$.
- $Y_i^{\text{obs}} = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ is the observed outcome when SUTVA and consistency hold.
- $\tau_i = Y_i(1) - Y_i(0)$ is unit i's treatment effect.

Estimands and estimators

- $\overline{Y}(z) = N^{-1} \sum_{i=1}^{N} Y_i(z)$ is the finite-population mean potential outcome under treatment level z.
- $\tau = N^{-1} \sum_{i=1}^{N} \tau_i = \overline{Y}(1) - \overline{Y}(0)$ is the finite-population average treatment effect. Qualify population and conditional versions with a subscript or argument, such as τ_P, $\tau(x)$, or τ_g.

- A hat denotes an estimator, a random function of the assignment and observed outcomes. Its value at the realized data is an estimate.
- $\hat{\tau}_{\mathrm{DM}}$ denotes the treated-minus-control difference in observed means.
- $\hat{\mu}_{z,\mathrm{HT}}$ and $\hat{\mu}_{z,\mathrm{H}}$ denote, respectively, the Horvitz–Thompson and Hájek estimators of $\overline{Y}(z)$. The corresponding treatment-effect estimators are $\hat{\tau}_{\mathrm{HT}} = \hat{\mu}_{1,\mathrm{HT}} - \hat{\mu}_{0,\mathrm{HT}}$ and $\hat{\tau}_{\mathrm{H}} = \hat{\mu}_{1,\mathrm{H}} - \hat{\mu}_{0,\mathrm{H}}$.
- $\hat{\mu}_{\lambda}$ denotes a member of the Trotter–Tukey affine normalization family, with Horvitz–Thompson at $\lambda = 0$ and Hájek at $\lambda = 1$. $\hat{\mu}_{\mathrm{AN}}$ denotes the adaptively normalized fixed-point estimator.
- $\overline{X} = N^{-1} \sum_{i=1}^{N} X_i$ is the finite-population covariate mean. Center covariates at $\overline{X}$ when writing regression-adjusted treatment effects.
- $\hat{\tau}_X = \overline{X}_1^{\mathrm{obs}} - \overline{X}_0^{\mathrm{obs}}$ denotes realized covariate mean imbalance.
- $m_z(x)$ denotes an outcome prediction for treatment arm z; distinguish a function fixed before assignment from one fitted using realized outcomes.
- $\hat{\mu}_{z,\mathrm{AIPW}}$ denotes the prediction-plus-IPW-correction estimator of $\overline{Y}(z)$, and $\hat{\tau}_{\mathrm{AIPW}} = \hat{\mu}_{1,\mathrm{AIPW}} - \hat{\mu}_{0,\mathrm{AIPW}}$.

Assignment and sampling probabilities

- $\mathbb{P}_Z$, $\mathbb{E}_Z$, Var_Z, and Cov_Z refer to probability under the assignment design, holding the finite population and all potential outcomes fixed.
- $\pi_i = \mathbb{P}_Z(Z_i = 1)$ is unit i's marginal treatment probability.
- When both treatment states appear symmetrically, use $\pi_i(z) = \mathbb{P}_Z(Z_i = z)$, so that $\pi_i(1) = \pi_i$ and $\pi_i(0) = 1 - \pi_i$.
- Use p for a common Bernoulli assignment probability.

Finite-population and superpopulation operators

- For $z \in \{0, 1\}$, $S_z^2 = (N-1)^{-1} \sum_{i=1}^{N} \{Y_i(z) - \overline{Y}(z)\}^2$ is the finite-population variance of potential outcomes.
- $S_\tau^2 = (N-1)^{-1} \sum_{i=1}^{N} (\tau_i - \tau)^2$ is the finite-population variance of unit-level treatment effects.
- Within stratum h, use τ_h, S_{zh}^2, and $S_{\tau h}^2$ for the stratum ATE and the corresponding finite-population variances.
- s_z^2 is the usual sample variance of observed outcomes within realized arm z.
- Unsubscripted probabilistic expectations such as $\mathbb{E}$ refer to a stated superpopulation or statistical model, not to the assignment design.

Networks and interference

Panel data and matrices

Probability, convergence, and norms

- $\mathbf{1}\{A\}$ is the indicator of event A. When a vector of ones is needed, write $\mathbf{1}_N$ for the length-N vector so it is not confused with the indicator notation.
- Always subscript probability, expectation, and variance when more than one source of randomness is in play.

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Problem Sets

The two course problem sets are completed individually.

Problem set	Assignment	Deadline
Problem Set 1	TBD	TBD
Problem Set 2	TBD	TBD

Projects

The three course projects are completed in groups of 2-4 students. Each project has three parts: a **proposal 1-pager**, an **in-class presentation**, and a **write-up**.

The proposal 1-pager is a one-page PDF uploaded to Canvas. It should state the project topic—including the paper selected for Project 1 or Project 2—list the group members, and give a one-paragraph summary of what the group intends to investigate.

Both proposal 1-pagers and write-ups are due at **11:59 p.m. ET** on the dates shown below.

Project	Proposal 1-pager due	In-class presentation	Write-up due
Project 1	September 11, 2026	Sep 30, 2026	Oct 7, 2026
Project 2	October 16, 2026	Nov 4, 2026	Nov 11, 2026
Project 3	November 20, 2026	Dec 9, 2026	Dec 16, 2026

Project 1

Choose a paper roughly corresponding to L01–L04, study its contribution closely, replicate its analysis, and consider possible extensions. The paper choice must be approved by the instructor before the group proceeds.

Possible papers

The following papers are starting points rather than a closed list. They span the material in L01–L04 and vary in theoretical, computational, and empirical emphasis. In the proposal 1-pager, identify a tractable result, simulation, or empirical analysis to replicate and a specific extension or sensitivity analysis to investigate.

- **Randomization inference beyond the sharp null (L01).** Caughey, Dafoe, Li, and Miratrix (2023), *Randomisation Inference beyond the Sharp Null: Bounded Null Hypotheses and Quantiles of Individual Treatment Effects*. Reproduce one of the proposed tests or bounds, then study its finite-sample behavior, computational demands, or sensitivity to the assignment design.
- **Estimated propensity scores and efficiency (L02).** Hirano, Imbens, and Ridder (2003), *Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score*. Reconstruct the paper's efficiency comparison and investigate when estimating rather than inserting a known propensity score helps in finite samples.

- **Adaptive normalization of inverse-probability weights (L02).** Khan and Ugander (2023), *Adaptive Normalization for IPW Estimation*. Replicate the comparisons among Horvitz–Thompson, Hájek, and adaptively normalized estimators, then alter the assignment design, outcome structure, or nuisance estimation method.
- **Regression adjustment in randomized experiments (L02–L03).** Lin (2013), *Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique*. Reproduce a central simulation or empirical comparison and examine the role of treatment–covariate interactions, imbalance, or unequal treatment allocation.
- **Rerandomization (L03).** Morgan and Rubin (2012), *Rerandomization to Improve Covariate Balance in Experiments*. Implement the design and reproduce its balance or precision gains, then vary the acceptance rule, balance metric, covariate dimension, or analysis procedure.
- **Algorithmic covariate balancing (L03).** Harshaw, Sävje, Spielman, and Zhang (2024), *Balancing Covariates in Randomized Experiments with the Gram–Schmidt Walk Design*. Implement a manageable part of the design and compare its balance–randomness tradeoff with complete randomization or rerandomization.
- **Limited overlap and optimal trimming (L04).** Crump, Hotz, Imbens, and Mitnik (2009), *Dealing with Limited Overlap in Estimation of Average Treatment Effects*. Reproduce the trimming rule and a key simulation, then examine how trimming changes both precision and the population whose effect is being estimated.
- **Smooth trimming and post-trimming inference (L04).** Yang and Ding (2018), *Asymptotic Inference of Causal Effects with Observational Studies Trimmed by the Estimated Propensity Scores*. Compare hard trimming with smooth weighting and study bootstrap inference or uncertainty from the design stage.
- **Heteroscedasticity-aware trimming (L04).** Khan and Ugander (2025), *Doubly Robust and Heteroscedasticity-Aware Sample Trimming for Causal Inference*. Replicate a simulation or application, then compare propensity-only and conditional-variance-aware trimming under alternative forms of heteroscedasticity.
- **Regularizing extrapolation (L04).** Arbour, Parikh, Niknam, Stuart, Rudolph, and Feller (2025), *Regularizing Extrapolation in Causal Inference*. Implement the proposed regularization and investigate the paper's bias–bias–variance tradeoff as overlap, dimension, or the outcome model changes.

Project 2

Choose a paper aligned with the material in L05–L07, study its contribution closely, replicate its analysis, and consider possible extensions. The paper choice must be approved by the instructor before the group proceeds.

Project 3: Open-ended research question

Propose a research question and conduct an open-ended initial investigation of it.

Further instructions for the presentations and write-ups will be provided separately.